Trade Shock and Environmental Spillover: The Causal Link Between U.S.-China Trade War and Increased Soy-Beef Driven Deforestation in the Brazilian Amazon.

Master's Thesis

of

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Abstract

This thesis investigates whether the 2018 U.S.–China trade war causally accelerated deforestation in the Brazilian Amazon. Retaliatory Chinese tariffs on U.S. agricultural products triggered a massive redirection of soybean and beef demand toward Brazil, generating an exogenous surge in Brazilian export volumes. To isolate this effect from endogeneity arising from domestic productivity gains and reverse causality, the study combines three complementary econometric strategies on high-frequency panel data (2012–2024): (i) baseline OLS regressions linking monthly export values to INPE DETER deforestation alerts, (ii) two-stage least squares (2SLS) instrumental-variable estimation that instruments Brazilian soybean exports to China with the U.S.–China Tension (UCT) Index, and (iii) a difference-in-differences design comparing Brazil (treatment) against the United States and Argentina (controls) before and after the 2018 shock.

Both IV and DiD specifications confirm a robust causal relationship: the trade-war-induced increase in Brazilian exports to China produced a statistically and economically significant rise in Amazonian forest loss. Placebo tests using exports to the European Union (unaffected by the shock) and parallel analysis of beef trade reinforce the identification strategy. The findings reveal that geopolitical tensions generate substantial indirect environmental externalities through global supply chains. They underscore the need for integrated trade and environmental policies—such as the EU Deforestation Regulation—that explicitly account for trade-diversion effects and prevent leakage to less-regulated markets.

1 Introduction

The Amazon rainforest, the world's largest tropical forest ecosystem, serves as a cornerstone of global climate regulation, biodiversity conservation, and regional hydrological stability. Despite its ecological criticality, Brazilian deforestation has accelerated over the past two decades, catalyzed primarily by the frontiers of agricultural expansion. Within this transformation, international trade—specifically the burgeoning global demand for commodities such as **soybeans and beef**—has emerged as a primary driver of land-use change. For policymakers, reconciling economic development with ecological preservation requires a rigorous understanding of the causal mechanisms through which global market shocks translate into local environmental degradation.

A pivotal disruption to these global trade dynamics occurred with the onset of the **2018 U.S.–China trade war**. Initiated by a rapid escalation of bilateral tariffs between the world's two largest economies, the conflict fundamentally reoriented global commodity flows. When China implemented retaliatory tariffs on U.S. soybeans, the resulting trade diversion shifted Chinese demand dramatically toward Brazilian suppliers. This policy-induced surge in Brazilian export volumes provides a compelling quasi-natural experiment to evaluate the environmental repercussions of large-scale, exogenous demand shocks.

This paper investigates whether **the U.S.–China trade war exerted a causal influence on Amazonian deforestation**. Establishing this link is methodologically complex due to inherent endogeneity: while trade demand may drive land clearing, increased agricultural capacity can also facilitate export growth, and simultaneous domestic policy shifts often confound simple correlations. Without isolating the exogenous component of this trade surge, empirical analyses risk misrepresenting the magnitude and nature of trade's environmental impact.

To resolve these endogeneity concerns, this research employs a causal inference framework utilizing three distinct but complementary econometric strategies. First, a baseline **Ordinary Least Squares (OLS) model** characterizes the correlation between export intensity and forest loss. Second, a **Two-Stage Least Squares (2SLS) Instrumental Variable (IV)** model is utilized to isolate exogenous variation in Brazilian trade; this approach leverages the **U.S.–China Tension (UCT) Index**, as developed by Rogers, Sun, and Sun (2024), as a plausibly exogenous instrument for redirected demand. Third, a **Difference-in-Differences (DiD)** specification compares the longitudinal trajectories of exports and deforestation in Brazil against those of the United States and Argentina—the primary counterfactual producers—to exploit the sharp structural break in trade flows occurring after 2018.

The 2018 shock occurred at a time when Brazil's soybean exports were already expanding, yet the trade war induced a surge nearly 40% above previous peaks. This redirection likely activated latent agricultural capacity, incentivizing the rapid conversion of primary forest and Cerrado land into cropland. By utilizing high-frequency satellite alert data from **INPE's DETER system**, this analysis examines the temporal and spatial granularity of this transition to determine if the export shock translated into measurable increases in forest clearing.

The empirical results presented in this paper reveal a consistent and statistically significant pattern: the trade war-induced escalation of Brazilian exports to China caused a measurable increase in

Amazonian deforestation. While OLS estimates demonstrate a positive association, the IV and DiD frameworks confirm a causal relationship. Specifically, the IV estimates—instrumented by the UCT Index—uncover an economically meaningful impact of geopolitical friction on tropical land use. Furthermore, the DiD analysis validates these findings by demonstrating that post-2018 deforestation trends in Brazil diverged sharply from the trends observed in its global competitors.

Ultimately, these findings suggest that international political tensions generate profound, albeit indirect, environmental externalities. While trade measures are typically designed to satisfy strategic or economic objectives, their ecological footprints are frequently undervalued in the policy discourse. By quantifying how geopolitical volatility reverberates through supply chains to accelerate deforestation, this paper emphasizes the urgent need for integrated trade and environmental policies that account for the globalized drivers of ecological decline.

2 Literature Review

Brazil's extensive forested regions, particularly the Amazon and the Cerrado, have long been central to global environmental discussions due to persistent and increasing deforestation. In the aftermath of the 2018 U.S.-China trade war, significant shifts in international trade flows—notably concerning soybeans—have intensified pressures on these ecosystems.

One key factor shaping this landscape is the advancement in agricultural technologies, particularly the widespread adoption of **genetically engineered (GE) soy seeds**. These innovations have considerably improved yields in Brazil. (Carreira, Costa, and Pessoa, 2024) quantifies the relative local impact of agricultural productivity gains and the rise in international demand on the expansion of agricultural land over forested areas. Specifically, they disentangle the effects of two quasi-exogenous indicators of local exposure to new genetically engineered (GE) soy seeds and increased demand from China on deforestation and land use across Brazilian municipalities from 2000 to 2017. They find that municipalities with greater exposure to GE soy varieties have experienced more rapid expansion of cropland and higher rates of deforestation. This dynamic reflects a version of the Jevons Paradox: as technological efficiency improves, the overall consumption of resources can still increase—in this case, via **land use expansion**.

Brazil's capacity to swiftly ramp up soybean production in response to heightened Chinese demand during the trade war may have been, in part, facilitated by earlier investments in these agricultural technologies. In this context, our instrumental variable (IV) approach, which identifies the causal impact of **geopolitical tensions on deforestation**, is well suited to quantifying the environmental consequences of this otherwise latent production capacity being activated by an external shock.

Interestingly, findings from other regions suggest that the relationship between productivity and deforestation is not universal. (Abman and Carney, 2020), studying agricultural programs in Malawi and Uganda, observed reductions in forest pressure where **productivity-enhancing interventions** were implemented. This comparison underlines the importance of contextual factors—such as market access and global demand—in shaping how productivity affects land use. In Brazil, where export demand is strong and infrastructure supports rapid scale-up, the link between **productivity** and **deforestation** appears much stronger.

Our paper focuses on the **exogenous disruption** caused by the 2018 U.S.-China trade war. As China imposed retaliatory tariffs on U.S. soybeans, it redirected its imports to Brazil. (Carreira et al., 2024) found that while there is **no direct correlation** between proximity to Chinese demand and deforestation, Chinese demand seems to moderate the deforestation-inducing effects of GE soy, indicating a more complex relationship between trade and land use change.

This nuance is crucial. It challenges simplistic narratives that attribute deforestation directly to trade increases. Instead, the **environmental impact** of trade depends on how and where agricultural expansion occurs. Our IV model uses the U.S.-China Tension Index (UCT Index) as a plausibly exogenous instrument to **isolate the effects of redirected demand** on Brazilian deforestation, allowing for more robust **causal inference**.

Our paper also contributes to the empirical literature on the **environmental implications of trade** (Copeland et al., 2021; Cherniwchan and Taylor, 2022). A series of cross-country and correlational studies have documented that **trade liberalization** is associated with **higher deforestation**. Abman and Lundberg (2020) show that regional trade agreements lead to a subsequent increase in deforestation, particularly in tropical regions. Additionally, several studies link increases in international commodity prices with deforestation in Brazil (Assunção et al., 2015; Harding et al., 2021; Da Mata and Dotta, 2021). Others have used related shift-share shocks to explore the effects of greater trade demand for exports, focusing on air pollution and emissions in China and India (Bombardini and Li, 2020; Barrows and Ollivier, 2021). A growing literature incorporates environmental externalities into trade models with **heterogeneous productivity** (Costinot et al., 2016; Shapiro, 2016; Hsiao, 2021; Pellegrina, 2022; Dominguez-Iino, 2021; Farrokhi et al., 2023).

Another critical dimension involves the **regulatory responses** to deforestation, particularly from the **European Union**. In 2023, the EU introduced the **Regulation on Deforestation-Free Products (EUDR)**, requiring importers to prove that their products—including **soy and beef**—are not linked to post-2020 deforestation. The regulation mandates detailed due diligence and geolocation data to trace supply chains.

The study "Trade Agreements, Import Bans, and the Leverage of the EU on Reducing Deforestation in Brazil" (2024) offers valuable insight into this policy environment. While the EUDR has been welcomed by environmental groups, concerns persist regarding its enforcement and potential impact on smallholders. Moreover, during the U.S.-China trade disruption, Brazil's primary export destination was China, which does not enforce similar environmental standards. This asymmetry in regulatory pressure is significant: while the EU may act to limit deforestation through policy, China's demand—absent such regulations—can exert the opposite effect.

This dynamic underscores the need to consider the **interdependence of trade and technology**. Treating them as isolated variables risks overlooking the real-world complexity of deforestation drivers. (Carreira et al., 2024) argue that while GE soy adoption is associated with increased deforestation, this effect is moderated in regions with high Chinese demand, possibly because expansion takes place in areas already cleared or more suited for agriculture.

Our empirical model is designed to test precisely this kind of relationship. By leveraging the **UCT Index**, we isolate the geopolitical shock component of increased soybean trade, allowing us to assess whether trade-induced expansion correlates with measurable increases in deforestation.

This methodology helps overcome common problems of **endogeneity** tied to commodity prices or internal productivity trends.

3 Background and History

3.1 Soy Deforestation and the Global Impact of Tariffs

The expansion of **soybean cultivation** has become one of the most significant drivers of **deforestation in Brazil**, with profound implications for both local ecosystems and global climate stability. The Amazon rainforest and the Cerrado savannah—two of the world's most biodiverse and ecologically vital regions—have been increasingly cleared to make way for **soybean production**. This expansion is particularly striking given that Brazil has over 25 million hectares of degraded or abandoned land suitable for cultivation, meaning that much of the forest loss associated with soy production is unnecessary.

China's growing appetite for soy has been central to this process. Since 2013, Chinese imports from Brazil have expanded rapidly, accelerating during the U.S.–China trade war when tariffs diverted demand away from U.S. suppliers. By 2020, Chinese soybean imports were associated with an estimated 229,000 hectares of deforestation—more than double Brazil's own domestic consumption of 102,000 hectares. On a per-tonne basis, China ranked second globally in deforestation exposure, just behind Portugal and slightly ahead of Brazil itself (Charbeneau, 2021). With over 75% of global soybean output destined for animal feed, rising meat and dairy consumption continues to link dietary change to **deforestation pressures**.

Tariffs have amplified these dynamics by reshaping global trade flows. The soybean market is unusually concentrated: China consumes nearly two-thirds of global imports, while the United States and Brazil supply most exports. When tariffs disrupted this balance, Brazil became China's default supplier, and researchers estimate that production could rise up to 39% above baseline levels as a result. For U.S. farmers, the consequences were immediate—lost sales and a $7.7 billion government relief package—while in Brazil the costs have been ecological. Fuchs (2019) projects that tariff-driven expansion could trigger land clearance at a pace four times greater than Brazil's peak deforestation year of 2004.

The consequences of this expansion extend beyond land use. The Amazon, often referred to as the "lungs of the planet," plays a vital role in absorbing carbon dioxide. Large-scale forest loss risks releasing massive carbon stores, undermining global climate targets. As Thomas Lovejoy has argued, "efforts to mitigate climate change cannot succeed if there is a substantial carbon release from the Amazon." Biodiversity loss presents an equally grave concern: the Amazon harbors species with untapped scientific and medical potential, as past pharmaceutical breakthroughs—such as ACE inhibitors developed from the venom of the Brazilian pit viper—demonstrate.

Policy interventions have sought to curb soy-driven deforestation, but with limited success. The **2006 Soy Moratorium** prohibited cultivation on newly deforested Amazon land, slowing direct conversion but failing to prevent indirect displacement, as land for soy expansion often came from pastures pushed deeper into the forest. Recent political shifts in Brazil have compounded

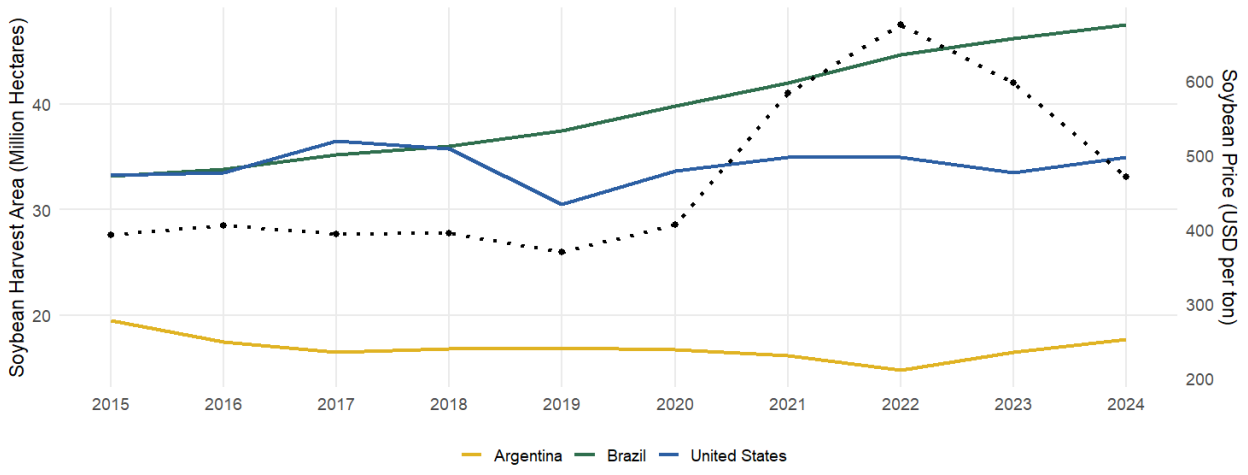

Figure 1: **Soybean harvest area in the three largest producing countries and global soybean price, 2015–2024**.
This figure shows the annual soybean harvested area (left axis, million hectares) for Argentina, Brazil, and the United States together with the average world soybean price (right axis, USD per ton) over the period 2015–2024. Brazil's harvested area expanded steadily from approximately 35 million hectares in 2015 to over 45 million hectares in 2024, while the United States experienced a temporary contraction around 2019 before stabilizing near 35 million hectares. Argentina's area remained relatively stable between 15 and 20 million hectares. Global soybean prices were flat until 2021, then rose sharply to a peak above 600 USD per ton in 2022 before declining again. These trends provide important context for the expansion of Brazilian soybean production and its potential links to land-use change examined in the empirical analysis. (Source: Own illustration)

these risks. Under President Jair Bolsonaro, enforcement of environmental protections weakened sharply, with pro-business rhetoric prioritizing economic growth over conservation. Critics warn that these policies accelerated clearance, though some large agribusinesses appear increasingly aware that continued deforestation risks exclusion from key markets, particularly in Europe, where sustainability standards are tightening.

In sum, the intersection of tariffs, trade realignment, and global demand has deepened the link between soybean expansion and deforestation. While U.S. farmers suffered economically, Brazil's ecosystems bore the ecological cost. The unintended consequences—accelerated forest loss, carbon emissions, and biodiversity decline—highlight the difficulty of reconciling economic competition with ecological sustainability in a globalized agricultural system.

3.2 The U.S.-China Trade Relationship

Over the past several decades, the trade relationship between the United States and China has undergone a significant transformation, evolving into one of the most consequential bilateral economic partnerships in the global economy. Since China's accession to the World Trade Organization (WTO) in 2001, **bilateral trade volumes** have expanded markedly. China has emerged as one of the **largest purchasers** of U.S. goods and services, ranking second only to Mexico, while the United States remains China's single most important export market. This intensification of trade has delivered tangible benefits for U.S. consumers through reduced prices and increased product variety, while simultaneously enhancing profit margins for American multinational corporations.

However, the expansion of this economic relationship has not been without adverse effects. The influx of Chinese imports, coupled with rapid automation and the relocation of manufacturing to low-cost regions, has contributed to widespread industrial decline in certain sectors of the U.S. economy. The resulting job displacements and wage pressures have become central themes in domestic political discourse surrounding globalization and trade policy.

Tensions in the bilateral relationship **escalated sharply in 2018**, when the Trump administration initiated a series of protectionist measures under its "America First" agenda, triggering what became widely known as the U.S.-China trade war. This marked a pivotal shift in U.S. trade policy, with the imposition of sweeping tariffs on Chinese imports. Despite diplomatic warnings from Beijing regarding the mutually detrimental nature of **tariff escalation**, many of these measures were subsequently maintained under the Biden administration, signaling a degree of bipartisan consensus on adopting a more confrontational economic stance toward China.

As of 2023, the United States and China collectively accounted for an estimated 43% of global gross domestic product (GDP) and approximately 48% of global manufacturing output, according to data from the World Bank. These figures underscore the central role that both economies play in shaping global trade patterns and industrial supply chains.

Despite growing geopolitical competition, the level of economic interdependence remains profound. In 2024, China ranked as the third-largest destination for U.S. exports, behind only Canada and Mexico, with American goods and services shipped to China valued at over 195 billion. Meanwhile, the **U.S. trade deficit** with China, although reduced to $295 billion, the lowest since 2009, remains the most significant among all of its trading partners. On the import side, Chinese goods constituted approximately 13.5% of total U.S. imports, just trailing Mexico. These imports are heavily concentrated in high-tech sectors, including electronic components, batteries, and computer hardware.

Additionally, financial linkages between the two countries further complicate the bilateral relationship. As of 2024, China held approximately $760 billion in U.S. Treasury securities, making it the second-largest foreign creditor to the United States, after Japan. This level of cross-border investment illustrates the deep structural entanglement of U.S.-China economic relations—where trade flows, supply chains, and financial markets remain tightly interconnected, despite rising **strategic tensions**.

Historically, however, this relationship developed gradually. After the establishment of the People's Republic of China in 1949, there was virtually no trade between the two countries for three decades, as the United States severed ties with Beijing's communist government. Relations began to shift in the late 1970s when China, under Deng Xiaoping's leadership, embarked on a long-term program of economic reform, loosening state control and encouraging private enterprise. In 1979, the United States and China normalized diplomatic and trade relations, with Beijing actively seeking foreign investment and market access. By 1986, China had formally applied to rejoin the **General Agreement on Tariffs and Trade (GATT)**, the precursor to the WTO (Anshu Siripurapu, Noah Berman, 2024).

China's formal accession to the WTO in December 2001 followed years of intensive negotiations and commitments to market liberalization, including significant tariff reductions, stronger protec-

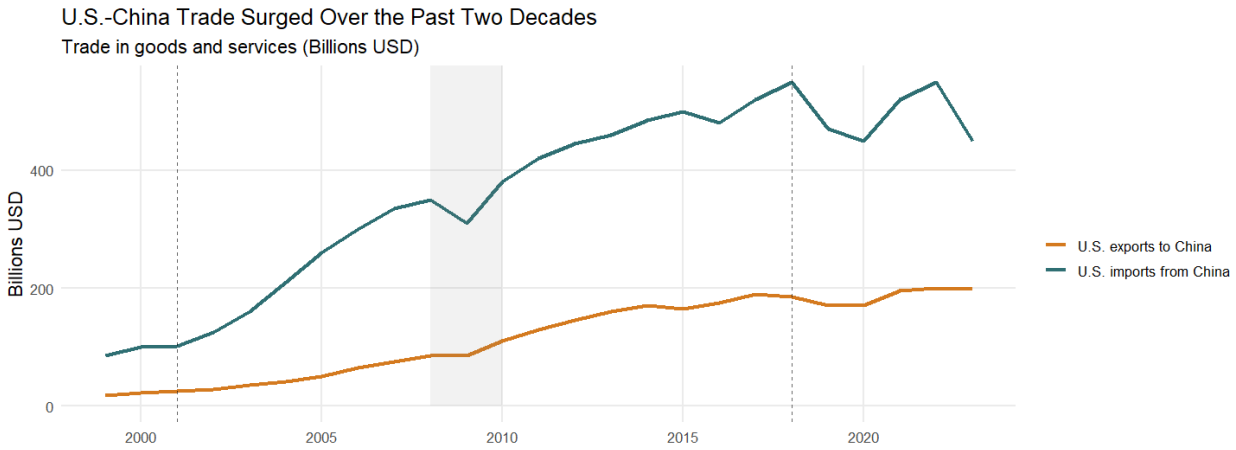

Figure 2: **U.S.-China bilateral trade in goods and services, 2000–2023**.
This figure shows U.S. exports to China (orange line) and U.S. imports from China (teal line) in billions of USD over the period 2000–2023. U.S. imports from China rose dramatically from approximately 100 billion USD in 2000 to a peak above 500 billion USD around 2018, while U.S. exports grew more modestly from about 20 billion USD to roughly 200 billion USD. The shaded area marks the Global Financial Crisis period (circa 2008–2010), and the vertical dashed line indicates the onset of the U.S.-China trade war in 2018. These trends underscore the widening bilateral trade imbalance and escalating geopolitical tensions, which motivated substitution toward alternative suppliers such as Brazil and provide essential context for the UCT_Index instrument and the empirical analysis of commodity-driven deforestation.
(Source: Own illustration based on "Council of Foreign Relations" chart found on: https://www.cfr.org/backgrounders/contentious-us-china-trade-relationship)

tions for intellectual property, and improved transparency in regulatory practices. The Clinton administration framed this development as a strategic opportunity—not only to open the Chinese market to U.S. exporters but also to potentially encourage political and institutional reform in China. Nonetheless, the decision provoked domestic backlash, particularly from labor unions and some members of Congress, who argued that China's lax labor and environmental standards would intensify a global "race to the bottom."

Following WTO accession, U.S.-China trade expanded at an unprecedented rate. Imports from China grew from approximately **$100 billion in 2001** to over **$400 billion by 2023**, driven by China's role as a critical hub in global value chains. Chinese factories often assemble goods using intermediate components sourced from multiple countries, making them essential to multinational supply networks. U.S. imports from China remain concentrated in sectors such as electronics, machinery, and nuclear reactors, while American exports to China are dominated by **agricultural commodities**—rendering them particularly vulnerable during trade disputes (Trading Economics).

Recurring tensions have persisted throughout the modern phase of the bilateral relationship, a pattern observable in the **U.S.-China Tension (UCT) Index** developed by John Rogers, Bo Sun, and Tony Sun (2024). Prior to 2018, the index showed periodic but limited volatility. However, following the implementation of **tariffs by the Trump administration in 2018**, the index began to **trend steadily upward**, indicating a period of sustained and systemic economic friction. This period of escalating trade conflict provides the empirical foundation for examining how global trade disruptions—particularly those involving China—have reshaped commodity flows, such as **soybean exports**, and their environmental consequences in third-party countries like **Brazil**.

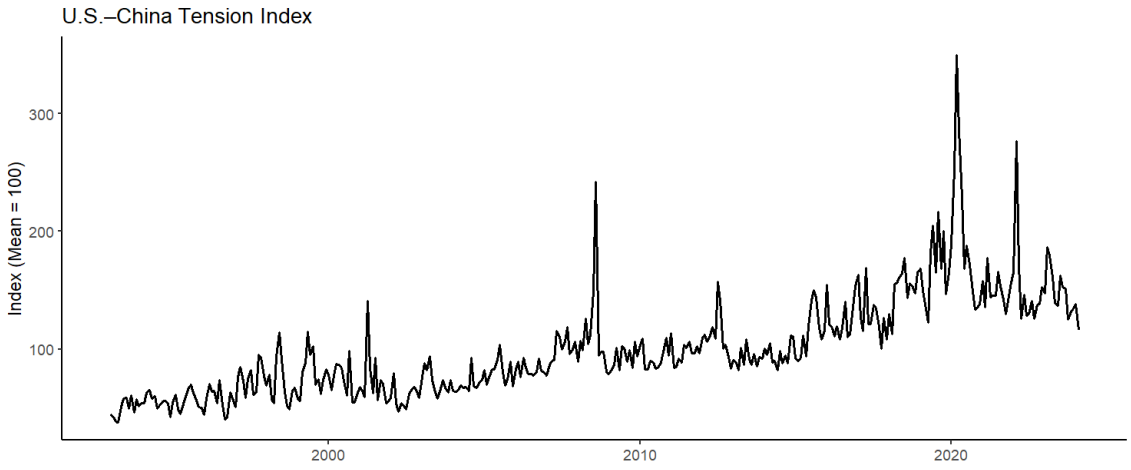

Figure 3: **U.S.–China Tension Index, 1995–2023**.
This figure presents the U.S.–China Tension Index (normalized so that the mean equals 100) over the period 1995–2023. The index, constructed from media-reported bilateral tensions, exhibits a clear upward trend with pronounced spikes corresponding to major geopolitical events. Key peaks are visible around 2008–2009, a dramatic surge during the 2018–2019 trade war (reaching over 350), and continued high volatility after 2020. This index serves as the exogenous instrumental variable (UCT_Index) in the 2SLS estimation. It captures unanticipated shifts in U.S.–China relations that strongly affect Chinese demand for soybeans and beef from the United States, inducing substitution toward Brazilian suppliers and thereby providing causal identification of the effect of commodity exports on deforestation in Brazil.
(Source: Own illustration based on US-China Tension Index found on:
https://www.policyuncertainty.com/US_China_Tension.html)

3.3 The U.S.-China Trade War (2018)

3.3.1 Early Escalation (2018)

In the early months of 2018, growing economic tensions and a shift toward **protectionist policy** in Washington laid the groundwork for **the U.S.-China trade war**. In January, the United States imposed tariffs on solar panels and washing machines, aimed at protecting domestic manufacturers. Two months later, new duties on steel and aluminum were introduced under the justification of national security. Although these global tariffs applied to all U.S. imports, China perceived the use of "national security" language as a troubling precedent and an implicit signal of more direct measures to come.

Simultaneously, the U.S. launched a broad investigation into Chinese trade practices. On March 22, 2018, U.S. Trade Representative Robert Lighthizer released a report accusing Beijing of engaging in unfair policies—particularly intellectual property theft, forced technology transfer, and industrial subsidies—costing the U.S. economy hundreds of billions of dollars annually. The following day, President Trump announced tariffs of 25% on $50 billion worth of Chinese imports, including aircraft components, batteries, medical equipment, and high-tech goods.

China's response was swift. On April 2, Beijing imposed **retaliatory tariffs** on 128 U.S. products—including pork, fruit, wine, and steel piping—designed to target politically sensitive sectors, particularly agriculture in Republican-leaning states. Though limited in scale, this move demonstrated Beijing's willingness to retaliate strategically.

On July 6, 2018, the first significant round of tariffs marked the official beginning of the trade war.

The U.S. implemented a 25% tariff on $34 billion of Chinese imports, primarily targeting sectors linked to China's "Made in China 2025" industrial strategy, while deliberately sparing most mass-market consumer goods such as cell phones, apparel, and toys. A second tranche covering $16 billion followed in August. China responded proportionally, imposing 25% tariffs on $34 billion of U.S. goods—including soybeans, beef, pork, and automobiles—and later on an additional $16 billion of imports. These reciprocal measures transformed the dispute into a structured, tit-for-tat trade conflict.

Although the initial tariff rounds were roughly symmetrical in dollar value, China faced constraints due to the large trade imbalance: in 2018, U.S. imports from China totaled $540 billion, compared with only $120 billion in U.S. exports to China. Beijing therefore had to target sectors that were both strategically significant and politically sensitive, such as agriculture and energy, rather than attempting to match tariffs dollar-for-dollar. Intensification and Stalemate (2019)

After a brief 90-day truce agreed at the G20 Summit in December 2018, negotiations resumed but soon stalled. In September 2019, the U.S. escalated by imposing tariffs on an additional 200 billion worth of Chinese goods, initially at 10% but later raised to 25% in May 2019. By year's end, U.S. tariffs covered approximately $360 billion of Chinese imports, while China had retaliated with duties on around $110 billion of American goods.

The broader economic effects were pronounced. Global supply chains began to adjust, with production shifting toward countries such as Vietnam and Mexico.

However, rather than large-scale reshoring to the U.S., firms pursued diversification strategies, underscoring the globalized nature of production networks. American consumers absorbed much of the tariff burden, while trade flows between the two countries contracted significantly.

The conflict also expanded beyond trade. In 2019, the U.S. blacklisted Huawei, China's leading telecommunications firm, on national security grounds, marking the first major escalation of the rivalry into the technology domain. China, in turn, began preparing countermeasures, including export restrictions on rare earth elements.

3.3.2 The Phase One Agreement (2020)

On January 15, 2020, after nearly two years of escalating tariffs, the two governments signed the Phase One agreement. The deal reflected mutual incentives to pause: U.S. markets were unsettled, farmers were increasingly vocal, and China's economy was slowing under tariff pressure.

The agreement focused on limited areas of compromise: • China pledged to purchase an additional $200 billion in U.S. goods and services over two years, primarily agricultural, energy, and manufactured products. • China agreed to strengthen intellectual property protections, ban forced technology transfers, and expand market access in financial services. • The U.S. cancelled some planned tariffs and reduced others, but retained 25% duties on roughly $250 billion worth of imports.

While the deal prevented further escalation, it did not address structural concerns such as Chinese industrial subsidies and the role of state-owned enterprises. The agreement institutionalized a more adversarial economic relationship, rather than resolving underlying disputes.

3.3.3 COVID-19 and Faltering Commitments (2020–2021)

The outbreak of COVID-19 weeks after the Phase One signing shifted priorities dramatically. Both countries took temporary steps to ease trade frictions in critical sectors: the U.S. suspended tariffs on certain medical supplies, while China reduced duties on pharmaceuticals and medical components.

Despite this limited cooperation, distrust deepened amid disputes over the pandemic's origins and early handling. Trade briefly contracted as global supply chains froze but rebounded by late 2020. By December, U.S. imports from China had returned to pre-trade-war levels, despite tariffs remaining in place.

Phase One quickly faltered. By the end of 2020, China had fulfilled only 57% of its first-year purchase commitments. While modest reforms were introduced in intellectual property and financial services, underperformance reinforced U.S. skepticism of negotiated concessions.

At the same time, Washington escalated its technology restrictions. The U.S. Commerce Department tightened export controls on Huawei and later blacklisted major firms including DJI (drones) and SMIC (semiconductors). China countered with its new Export Control Law and provisions for an Unreliable Entity List, followed in 2021 by the Anti-Foreign Sanctions Law. These developments marked a decisive shift toward technological decoupling.

3.3.4 WTO Blowback and Transition to the Biden Administration (2020–2021)

In September 2020, a WTO panel ruled that U.S. tariffs violated international trade rules, siding with China. However, Washington appealed to the WTO's incapacitated appellate body, effectively stalling the case. This episode underscored the broader risks the trade war posed to the multilateral trading system.

In its final weeks, the Trump administration passed the CHIPS for America Act, laying groundwork for domestic semiconductor support. When President Joe Biden took office in January 2021, expectations of a reset proved misplaced. The Biden administration maintained most tariffs, framing them as leverage to pressure Beijing on structural issues. U.S. Trade Representative Katherine Tai emphasized enforcement of Phase One commitments and greater alignment with allies, while launching federal reviews of supply chain vulnerabilities in semiconductors, critical minerals, pharmaceuticals, and batteries.

Human rights also entered the trade agenda. In December 2021, Congress passed the Uyghur Forced Labor Prevention Act (UFLPA), effectively banning imports from Xinjiang unless companies could prove absence of forced labor. This linked trade policy explicitly with values-based foreign policy, prompting sharp Chinese criticism.

By the end of 2021, the Phase One agreement had expired without renewal, and tariffs remained in place. Rather than easing, rivalry had intensified, with economic, technological, and human rights dimensions now deeply intertwined.

3.3.5 Supply Chain Realignment (2022–2024)

By 2022, structural adjustments to global trade flows became visible. China's share of U.S. goods imports fell from around 21–22% in 2017 to approximately 17%—its lowest since 2006. Meanwhile, imports from Mexico and Vietnam surged, with Mexico surpassing China as the U.S.'s largest source of imports in 2023.

Nevertheless, analysts noted that much of this reflected trade diversion rather than full-scale decoupling. Chinese firms increasingly established operations in Southeast Asia and Latin America, embedding Chinese components within goods re-exported to the U.S. Thus, while U.S. companies pursued "China plus one" strategies to reduce risk, underlying supply chain dependence persisted.

The U.S.-China trade war had profound implications not only for bilateral economic relations but also for **global trade dynamics**. Among the most significant shifts was the reorientation of China's **agricultural imports**. As tariffs on U.S. goods made American products less competitive, Brazil capitalized on this opportunity, emerging as a key supplier to China. However, this shift might have come with significant environmental costs. As demand for **Brazilian soybeans** soared, so did deforestation in the Amazon rainforest, driven by the expansion of agricultural land. The trade war thus may have had unintended global consequences, linking U.S.-China economic rivalry to ecological challenges in South America.

3.4 The United States, Brazil, and China Soybean Triangle

Over the last two decades, **Brazilian soybean exports** have expanded at an extraordinary pace, rising by 431% from 705 million bushels in 2004 to 3,744 million bushels in 2023. This surge has been particularly pronounced since the early 2010s, reflecting Brazil's emergence as a global agricultural powerhouse. Today, soybeans represent the country's most significant agricultural export by volume, accounting for over 60% of domestic production. According to Brazil's National Supply Company (Conab), the soybean harvest for the 2022/23 marketing year reached a record-breaking 5,680 million bushels.

For much of the twentieth century, the United States dominated the global soybean export market. However, in 2013 Brazil overtook the U.S. in shipments for the first time, a symbolic moment that foreshadowed a **long-term rebalancing of global soybean trade**. Since then, Brazil's market share has expanded steadily, and by 2023 its exports reached an unprecedented 3,744 million bushels—more than double the 1,789 million exported by the United States (Secex; USDA).

China has been the central driver of Brazil's rise. In 2012, China accounted for 70% of Brazil's soybean exports, a figure that peaked at 82% in 2018 during the height of the U.S.-China trade war. By 2023, 73% of Brazil's record shipments—amounting to 2,737 million bushels—were directed to China, representing a 40% increase over the previous year. This expansion was enabled by a bumper domestic harvest and competitive global pricing. Brazil's previous record had also coincided with a geopolitical shock: in 2018, when China sharply reduced purchases of U.S. soybeans in response to the imposition of 25% retaliatory tariffs, Brazilian exporters were well positioned to absorb the redirected demand.

China's role in the global soybean market remains decisive. Accounting for roughly **60% of world imports**, Chinese demand is largely driven by the need for animal feed and edible oils—essential

components of an increasingly diverse and affluent domestic diet. Historically, the United States served as China's main supplier, but over the past 15 years the center of gravity has shifted toward South America, particularly Brazil. Between 2019 and 2023, an average of 73% of Brazil's soybean exports went to China, compared to just 51% for the United States.

The U.S.-China trade war significantly disrupted established patterns. Between the mid-2000s and 2011, around **60% of U.S. soybean exports** were directed to China. However, this share collapsed to just **18% in 2018**, the nadir of the trade conflict. Although U.S. exports have partially recovered since 2020, averaging just above 50%, they remain below pre-trade war levels. American producers have sought to diversify their markets, but these efforts have not fully offset the decline in Chinese demand. In 2023, U.S. soybean exports to China fell by 12% compared to the previous year, amounting to 977 million bushels.

After nearly two decades of continuous growth, China's soybean imports have shown signs of plateauing since 2019. Several factors account for these fluctuations, including disruptions from the COVID-19 pandemic and outbreaks of African swine fever, which reduced feed demand for the pork industry. Looking forward, some analysts suggest that Chinese soybean imports may have already peaked. The country's current Five-Year Agricultural Plan (2021–2027) emphasizes boosting domestic soybean production to reduce reliance on foreign suppliers, a policy shift that could **reshape trade flows** between the United States, Brazil, and China (Colussi, Schnitkey, Janzen, & Paulson, 2024).

3.5 The European Union and the EUDR

The European Union Deforestation Regulation (EUDR) seeks to reduce deforestation in tropical producer countries by restricting the import of agricultural commodities, including soy. The main concern, however, is that such trade restrictions risk displacing deforestation-linked products toward unregulated markets rather than eliminating environmentally harmful practices altogether.

The EU livestock sector illustrates the stakes involved. Accounting for nearly 45% of total **EU agricultural output**, livestock production depends heavily on compound feed that relies on **imported soymeal**. Each year, approximately 150 million tons of compound feed are produced in the EU, requiring 28 million tons of soymeal—supplied either directly as meal or indirectly through whole soybeans processed in Europe (Tillie & Rodríguez-Cerezo, 2015). More than **70%** of this **soy** originates from **Brazil** and **Argentina** (Kroes & Kuepper, 2015; Kuepper & Stravens, 2022). This high level of dependency exposes the livestock sector to potential trade distortions, price volatility, and rising feed costs (de Boer et al., 2014).

Moreover, EU food safety and labeling rules add further complexity. Since 2003, EU regulations have required food and feed containing more than 0.9% genetically modified (GM) material to be labeled and traced (European Commission, 2003). This has effectively created segregated supply chains, and accidental traces of unauthorized GM material often lead to shipment rejections. Segregation costs, including contamination risks, have been estimated to increase feed costs by up to 300% (European Commission, 2007; Kalaitzandonakes et al., 2014).

The EUDR—Regulation (EU) 2023/1115, adopted in June 2023 and scheduled to come into effect in December 2025—represents a significant expansion of earlier EU timber legality standards.

Going beyond the EU Timber Regulation (EUTR), the EUDR includes soy alongside cattle, palm oil, cocoa, coffee, and rubber (European Union, 2023). The regulation prohibits the placing of products linked to deforestation or forest degradation on the EU market and requires companies to submit due diligence statements verifying compliance. These obligations include geospatial traceability, land tenure documentation, and adherence to local laws. Enforcement falls to member states, which must establish monitoring mechanisms, conduct compliance checks, and apply sanctions. Penalties can include fines of at least 4% of annual turnover, confiscation of goods or revenues, exclusion from public procurement, and temporary prohibition from accessing the EU market.

Soy is particularly significant within the EUDR framework. Four products are covered under Harmonized System (HS) codes: **HS 1201 (soybeans)**, HS 120810 (soymeal), HS 1507 (soy oil), and HS 2304 (oilcake). Of these, soybeans represent 64.91% of the trade value (USD 90.97 billion), followed by soybean oilcake (24.39%), soy oil (10.34%), and soymeal (0.36%). Major soy exporters to the EU—including **Brazil**, **Argentina**, Paraguay, and Bolivia—fall into the "standard risk" category, requiring member states to check at least 3% of companies sourcing from these countries annually.

While the EUDR demonstrates the EU's ambition to leverage trade policy for environmental ends, scholars caution that its effectiveness is limited by the dynamics of global commodity markets. Restrictions on South American exports are likely to divert trade toward alternative suppliers such as the United States and Canada, which may redirect soy exports to the EU under EUDR compliance, while Brazilian soy excluded from Europe is reallocated to China (Garrett et al., 2023; Colussi et al., 2025). Rather than inducing pro-forest land use change, these restrictions risk producing mainly **trade reallocation effects**, thereby undermining the regulation's environmental objectives. As South America grows increasingly reliant on Chinese demand, the EU's leverage to promote sustainable practices may diminish further.

4 Panel Data Analysis

To analyze the relationship between the U.S.–China trade conflict and Amazon deforestation, we constructed a **panel dataset** of soybean and beef export volumes from Brazil, the United States, and Argentina—the three largest soybean exporters to China. The year **2018** is treated as the "intervention" point, as it marks the onset of the U.S.–China trade war. Observing trade flows before and after 2018 allows us to identify disruptions in export patterns and their potential environmental consequences. Semi-annual export volume data were employed to capture seasonal variation and improve comparability, given that Brazil and the United States operate in opposite hemispheres with distinct agricultural cycles.

4.1 Analysis of Soybean Export Value to China

The panel data analysis reveals a **structural shock** to global soybean trade values following the onset of the U.S.–China trade conflict in 2018, providing the empirical foundation for the subsequent causal analysis. The accompanying figure illustrates the semi-annual value of soybean exports

to China, reported in thousands of U.S. dollars, for the three largest global suppliers: Brazil, the United States (U.S.), and Argentina.

Prior to 2018, exports from Brazil and the U.S. followed a predictable alternating seasonal pattern, reflecting their opposite-hemisphere harvest cycles. During these years, Brazil's export values typically peaked in the first half of the year (H1), reaching approximately 10 billion, while U.S. exports peaked in the second half (H2) at comparable levels. Argentina, though a consistent supplier, occupied a distant third position in export value throughout this period. The imposition of Chinese retaliatory tariffs on U.S. soybeans in July 2018 marked a decisive break in this pattern:

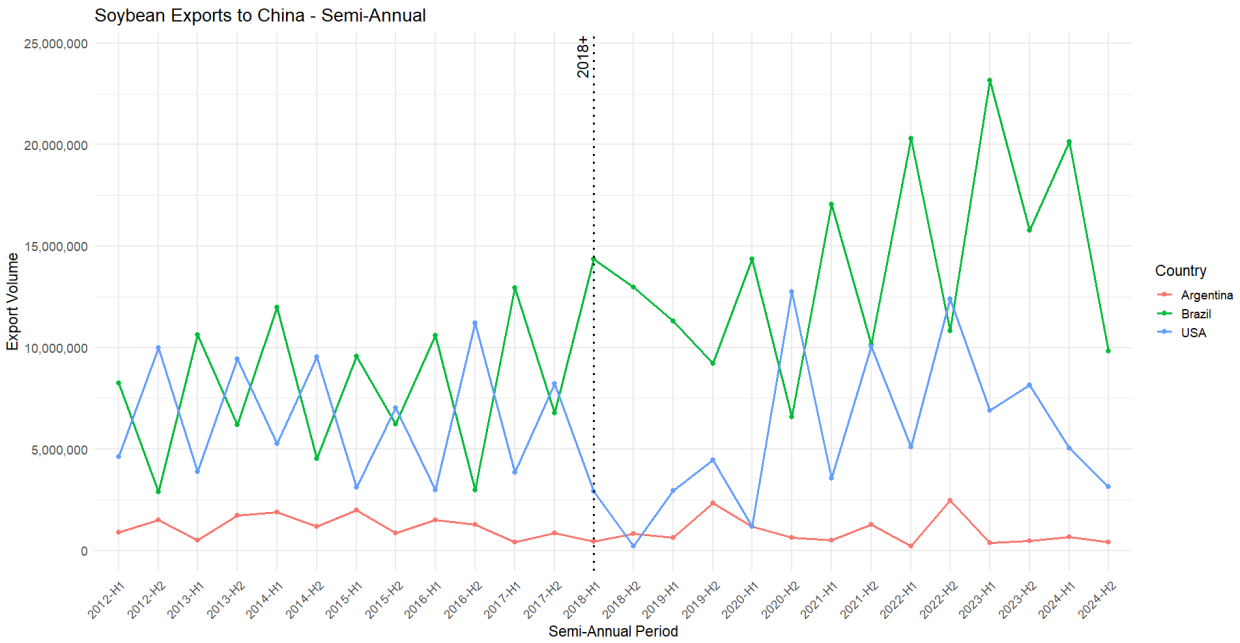

Figure 4: **Soybean exports to China by major suppliers, semi-annual, 2012–2024**. This figure displays semi-annual soybean export volumes to China from Argentina (red), Brazil (green), and the United States (blue) over the period 2012–2024. Prior to 2018, the United States and Brazil were broadly comparable suppliers, with export volumes fluctuating between 3 and 12 million tons per half-year. The vertical dashed line marks the intensification of the U.S.–China trade war in 2018. Following this shock, U.S. exports collapsed almost to zero while Brazilian exports surged dramatically, frequently exceeding 15 million tons and reaching peaks above 23 million tons per semi-annual period. Argentina's exports remained negligible throughout. This sharp substitution from the United States to Brazil illustrates the first-stage mechanism of the UCT_Index instrument and provides direct visual evidence of the exogenous increase in Brazilian soybean exports used to identify the causal effect on deforestation. (Source: Own illustration)

• **Brazilian Surge (Treatment Group)**: Brazil's export value to China **rose sharply**, reaching nearly **14 billion** in 2018-H2—an increase of roughly **40%** above its prior peak levels. This surge highlights Brazil's ability to absorb the redirected demand from China, reflecting both latent production capacity and the responsiveness of its export sector. In subsequent years, Brazil consistently exceeded its pre-2018 peaks, consolidating its position as China's leading soybean supplier.

• **U.S. Collapse (Control Group)**: In contrast, U.S. exports to China **plummeted** during its peak harvest season in 2018-H2, falling to just 1.5 billion. Remarkably, U.S. export values during this period were temporarily surpassed by Argentina, underscoring the severity of the trade shock.

While U.S. exports partially recovered after 2020, **they have not returned to pre-trade-war levels** and continue to trail Brazil by a wide margin.

• **Trade Diversion**: The redistribution of trade flows away from the U.S. toward Brazil reflects a classic trade diversion effect: an exogenous geopolitical shock that permanently restructured Chinese import patterns. This shift redirected billions of dollars in soybean trade value and established Brazil as the **dominant** supplier in the Chinese market.

This dramatic reallocation of export value represents the central "treatment shock" within the Difference-in-Differences (DiD) framework and the primary channel through which the U.S.–China trade war influenced land use dynamics in Brazil. The observed patterns also validate the use of the U.S.–China Tension (UCT) Index as an instrumental variable, since the policy-induced trade disruption cleanly isolates variation in Chinese demand for Brazilian soybeans. By linking this surge in export value to subsequent increases in deforestation pressure, we empirically assess the environmental consequences of the trade conflict.

4.2 Analysis of Soybean Export Value to the European Union

In contrast to the dramatic restructuring of trade flows observed in the Chinese market, the European Union (EU) provides a **benchmark of stability** in soybean trade dynamics. The semi-annual export values of soybeans to the EU from Brazil and the United States (U.S.), denominated in thousands of U.S. dollars, display a consistent seasonal pattern over the 2012–2024 period without evidence of a structural break following the onset of the U.S.–China trade war in 2018.

Several key patterns emerge:

• **Sustained Seasonality and Stability**: Both Brazil and the U.S. maintain relatively **stable seasonal cycles** throughout the sample period. Brazil remains the dominant supplier, with peak semi-annual export values regularly exceeding 2 billion, while U.S. exports fluctuate within a narrower band, typically between 750 million and 1.5 billion. Importantly, these patterns **persist after 2018**, suggesting an absence of diversionary effects comparable to those observed in China.

• **Minor EU Response to Trade War**: Unlike China, the EU did not impose retaliatory tariffs on U.S. soybeans and therefore did not experience a dramatic collapse in U.S. export values. Instead, the U.S. continued to supply the EU along its preexisting seasonal trajectory, with no evidence of the trade war-induced disruptions that characterized the Chinese market.

• **Regulatory Asymmetry and Control Group Function**: The EU's regulatory environment—shaped by stricter sustainability standards such as the forthcoming European Union Deforestation Regulation (EUDR)—distinguishes it from China, which has not imposed comparable environmental requirements on soybean imports. This divergence reinforces the EU's role as a control market in our empirical framework: its trade flows were largely insulated from the geopolitical shock of 2018, but remain subject to distinct regulatory constraints on supply chain sustainability.

The persistence of stable trade patterns in the EU market supports the identification strategy of the Difference-in-Differences (DiD) model. By contrasting the unaffected EU market with the shock-induced reorientation of Chinese imports, we isolate the exogenous impact of the U.S.–China trade

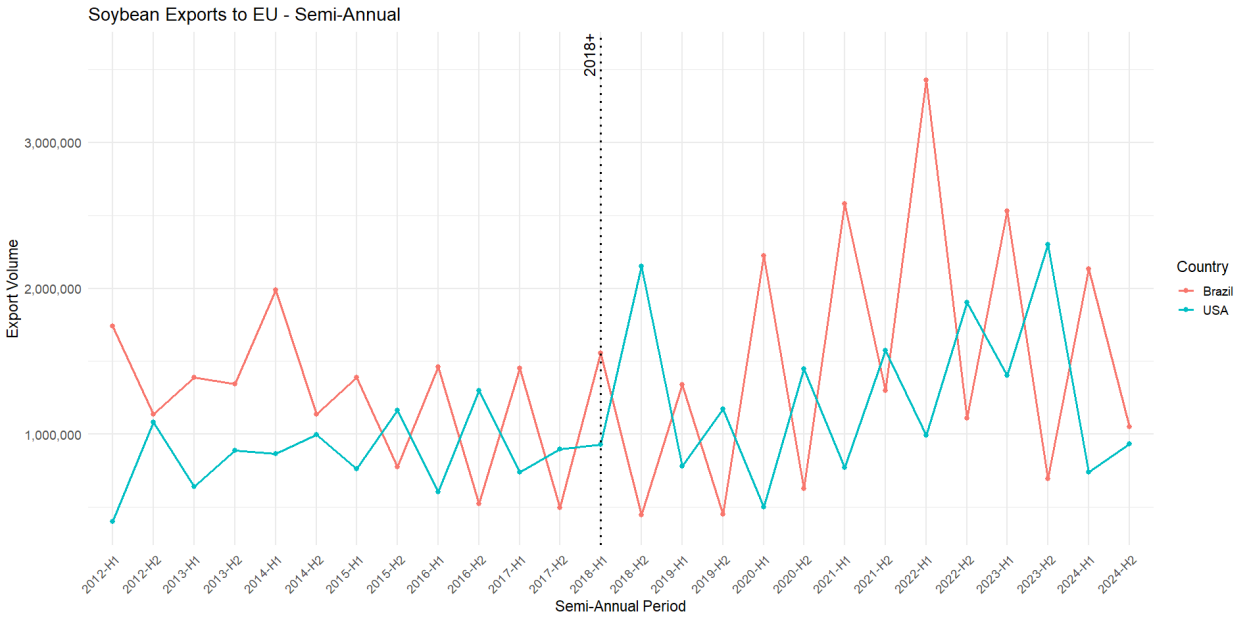

Figure 5: **Soybean exports to the EU by Brazil and the United States, semi-annual, 2012–2024**.

This figure displays semi-annual soybean export volumes to the European Union from Brazil (red line) and the United States (teal line) over the period 2012–2024. The vertical dashed line marks the intensification of the U.S.–China trade war in 2018. In sharp contrast to exports to China, there is no collapse in U.S. exports or dramatic reallocation toward Brazil after 2018. Both countries continued to supply the EU market, with volumes fluctuating between approximately 0.5 and 3.5 million tons per half-year; Brazilian exports show occasional large peaks (notably in 2021), while U.S. exports maintain a comparable range with spikes around 2018–2019 and 2022–2023. This stability in EU-bound exports demonstrates that the trade shock was specific to China, supporting the exogeneity of the UCT_Index instrument and validating the identification strategy used in the main analysis.

(Source: Own illustration)

war on Brazilian soybean exports. More broadly, this comparison underscores the thesis's central claim: **asymmetric regulatory environments shape the geographical locus of trade-induced land-use change, with China's laxer standards amplifying deforestation pressures in Brazil, while the EU's evolving regulations constrain such outcomes.**

4.3 Analysis of Beef Export Value to China

The analysis of semi-annual beef export values from Brazil to China reveals a clear and **sustained acceleration** in trade following the onset of the U.S.–China trade war in 2018. The figure, which reports export values in thousands of U.S. dollars, captures both the magnitude and persistence of this expansion, illustrating how Chinese demand for Brazilian beef became a structural feature of post-2018 trade dynamics.

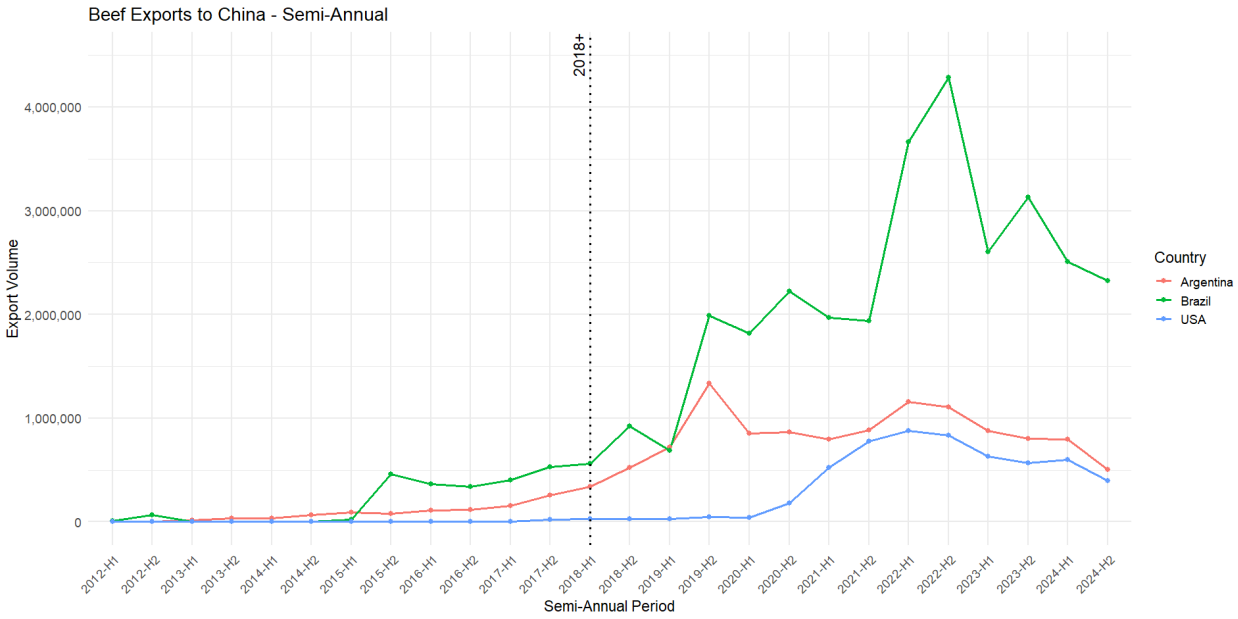

Figure 6: **Beef exports to China by major suppliers, semi-annual, 2012–2024**.
This figure displays semi-annual beef export volumes to China from Argentina (red), Brazil (green), and the United States (blue) over the period 2012–2024. The vertical dashed line marks the intensification of the U.S.–China trade war in 2018. Prior to 2018, beef exports to China were negligible across all three countries. Following the trade shock, Brazilian exports surged dramatically from near zero to peaks above 4 million tons per semi-annual period (notably in 2021–2022), while U.S. exports remained very low until a modest increase after 2020 and Argentine exports grew only modestly. This sharp reallocation toward Brazilian beef suppliers mirrors the soybean pattern and provides further evidence of the substitution mechanism induced by the UCT_Index instrument, reinforcing the causal identification of commodity export effects on deforestation in Brazil.
(Source: Own illustration)

• **Pre-2018 Trend**: Prior to 2018, Brazilian beef exports to China displayed moderate growth but remained volatile, typically fluctuating below 500 million per semi-annual period. The market, while expanding, was still **secondary** compared to soybean trade in terms of both value and strategic importance.

• **Post-2018** Acceleration: After the imposition of tariffs on U.S. agricultural products, Brazilian beef exports to China entered a phase of rapid and sustained growth, peaking near 2 billion per

semi-annual period by 2021–2022. This marked a fourfold increase over pre-trade-war levels and reflected China's pivot toward Brazil as its **principal supplier** of animal protein. The persistence of these elevated trade values underscores the structural nature of this realignment, rather than a temporary response to price fluctuations or supply constraints.

• **Reinforcement of the Soybean Findings**: Whereas the soybean trade exhibited an immediate and acute reallocation shock—characterized by Brazil replacing U.S. market share—the beef sector demonstrates a more gradual but sustained expansion of Chinese demand. Together, these trends indicate that the U.S.–China trade conflict triggered a broader reorientation of China's agricultural imports toward Brazil across multiple commodities. This pattern amplifies land-use pressures through both direct and indirect channels, contributing to Indirect Land Use Change (ILUC) as expanding pasture and feed crop production increasingly overlap in frontier regions.

In sum, the parallel expansion of Brazilian soy and beef exports to China highlights the **multifaceted nature** of the trade war's environmental consequences. The geopolitical disruption did not merely shift trade in a single commodity—it intensified China's overall reliance on Brazilian agricultural outputs, reinforcing the causal mechanism through which external demand shocks can drive deforestation and ecosystem degradation in the Amazon.

4.4 Analysis of Beef Export Value to the European Union

The analysis of semi-annual beef export values to the European Union (EU) provides an **essential counterfactual** to the sharp growth observed in the Chinese market. The figure, reporting export values in thousands of U.S. dollars, highlights a stable and predictable trade pattern throughout the 2012–2024 period, underscoring the EU's role as a regulated and steady market largely insulated from the geopolitical disruptions of the U.S.–China trade war.

• **Stable Pattern and Market Dominance**: The series displays a highly seasonal yet remarkably **stable pattern**, with no observable structural break or long-term upward trend following 2018. This stability indicates that the EU beef market remained unaffected by the trade diversion dynamics that reshaped agricultural trade flows between China and Brazil.

• **Regulatory Asymmetry and Market Constraints**: The absence of trade shocks in the EU market reflects its stringent regulatory environment. European imports are governed by rigorous sanitary and phytosanitary (SPS) standards, as well as emerging sustainability regulations such as the **European Union Deforestation Regulation (EUDR)**. These frameworks constrain the potential for sudden, large-scale expansions of supply, particularly from regions associated with deforestation risk. Consequently, the EU's regulatory safeguards function as a natural stabilizer, preventing the kind of unregulated trade surges observed in the Chinese market.

• **Robustness and Counterfactual Evidence**: The divergence between the EU and China markets mirrors the pattern observed in soybean trade. Whereas Chinese demand surged in response to trade war–induced supply shocks, EU trade remained stable and governed by pre-existing institutional constraints. This contrast reinforces the identification strategy of the empirical analysis: environmental pressures linked to agricultural expansion are concentrated in trade flows directed toward less regulated markets that experienced exogenous geopolitical shocks. The EU market, by

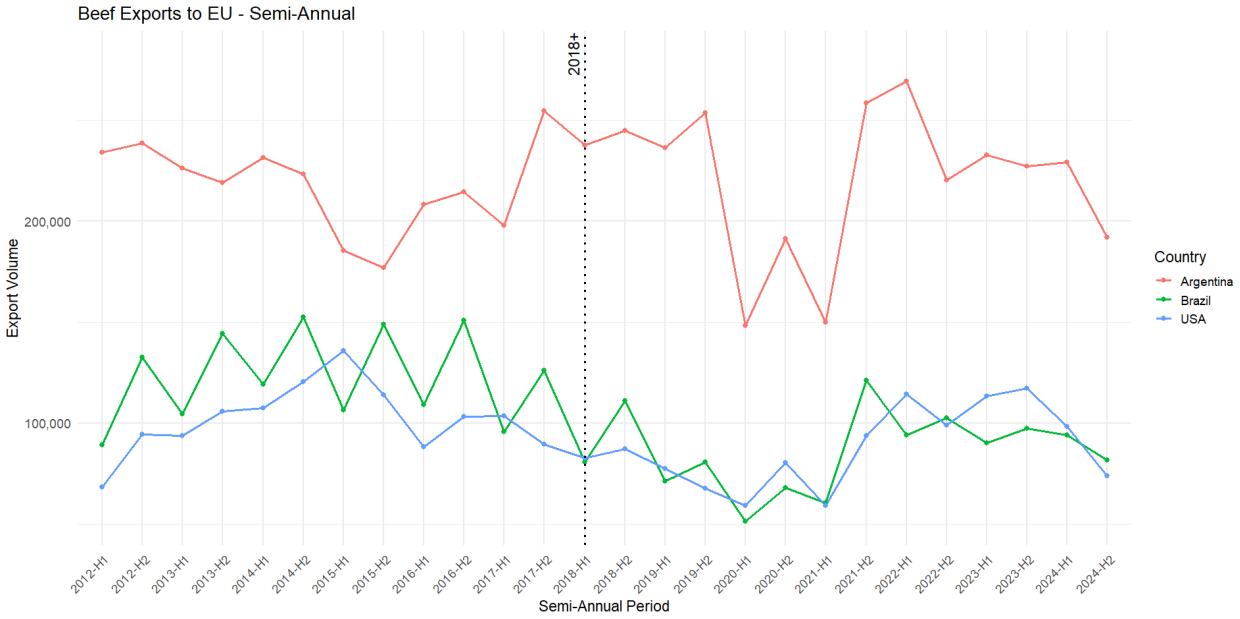

Figure 7: **Beef exports to the EU by major suppliers, semi-annual, 2012–2024**.
This figure displays semi-annual beef export volumes to the European Union from Argentina (red), Brazil (green), and the United States (blue) over the period 2012–2024. The vertical dashed line marks the intensification of the U.S.–China trade war in 2018. In clear contrast to exports to China, there is no collapse in U.S. exports or surge toward Brazil after 2018. Argentina remains the dominant supplier throughout, with volumes fluctuating between approximately 150,000 and 300,000 tons per half-year. Brazilian and U.S. exports stay much smaller (typically below 150,000 tons) and show no systematic reallocation or upward trend post-2018. This stability in EU-bound beef exports demonstrates that the trade shock was specific to the Chinese market, further validating the exogeneity of the UCT_Index instrument and reinforcing the causal identification strategy used in the empirical analysis.
(Source: Own illustration)

contrast, provides a credible counterfactual for isolating the causal effect of the U.S.–China trade war on Brazil's export behavior and associated land-use change.

5 Econometric Methodology

To establish a **causal link** between the 2018 U.S.–China trade war and deforestation dynamics in Brazil, this study employs a multi-method econometric strategy combining three complementary regression approaches: **Ordinary Least Squares (OLS), Instrumental Variables (IV), and Difference-in-Differences (DiD)**. Together, these models create a coherent empirical framework in which the OLS identifies baseline associations, the IV corrects for reverse causality and omitted variable bias, and the DiD validates the exogeneity of the trade shock underlying the causal inference.

Our empirical strategy identifies causal linkages between trade shifts and deforestation. We implemented three complementary econometric approaches:

5.1 Ordinary Least Squares (OLS) Regression

The initial step employs an **OLS regression** to examine the statistical relationship between international trade flows and monthly Amazonian deforestation alerts. The model incorporates two principal variables of interest—Brazilian soybean exports to China and Brazilian beef exports to China—which represent the primary trade channels affected by the 2018 U.S.–China tariff escalation. To mitigate confounding effects and capture broader trade displacement patterns, the model also includes a comprehensive set of control variables.

The baseline time-series specification is estimated using Ordinary Least Squares (OLS) and is defined as:

$$\begin{aligned} Deforestation_t = \beta_0 + \beta_1 BrazilSoyExportsChina_t + \beta_2 USASoyExportsChina_t \\ + \beta_3 ArgentinaSoyExportsChina_t + \beta_4 BrazilSoyExportsEU_t \\ + \beta_5 BrazilBeefExportsChina_t + \beta_6 USABeefExportsChina_t + \varepsilon_t \end{aligned} \tag{1}$$

Independent variables include:

Variables of Interest: Brazilian Soy Exports to China, Brazilian Beef Exports to China

Trade War Controls: U.S. Soy Exports to China, Argentine Soy Exports to China, U.S. Beef Exports to China (These variables control for market substitution and the displacement of trade volume due to tariffs.)

Control Market Variable: Brazilian Soy Exports to the European Union (Serves as a benchmark for trade with a highly regulated and stable market unaffected by the 2018 shock.)

The dependent variable is monthly deforestation alert area, measured in square kilometers and derived from INPE's Real-Time Deforestation Detection System (DETER).

All trade variables are measured in monthly export values, denominated in thousands of U.S. dollars.

5.2 Instrumental Variable (IV) Regression Design

To address endogeneity concerns, we employ a **Two-Stage Least Squares (2SLS) IV regression**, using the U.S.–China Tension (UCT) Index as an exogenous instrument for Brazilian soybean exports to China. The UCT Index, developed by Rogers, Sun, and Sun (2024), provides an exogenous source of variation suitable for identifying causal effects in international trade. This index quantifies tension between the United States and China by tracking related coverage in leading U.S. newspapers, following the methodology of Baker, Bloom, and Davis (2016) for measuring economic policy uncertainty. Specifically, it computes the share of articles that simultaneously reference (i) the United States (or U.S.) and China (or Chinese), (ii) contentious bilateral issues such as trade, technology, or security, and (iii) phrases indicating conflict or rivalry.

The construction of the UCT Index relies on advanced natural language processing techniques, including K-means clustering, guided Latent Dirichlet Allocation (LDA), and Newsmap analysis,

to identify relevant search terms from a large corpus of manually selected articles discussing U.S.–China tensions. The index is normalized to have an average value of 100 over the sample period from 1993 to 2024, allowing for consistent measurement of **tension fluctuations**. It has been validated against alternative sources, such as firm-level earnings call transcripts, U.S. legislative records, and documented episodes of U.S.–China trade and security conflicts, demonstrating its robustness in capturing genuine geopolitical frictions rather than noise.

The UCT Index satisfies the two essential IV assumptions: relevance and exclusion restriction, making it a strong and valid instrument in this context.

Relevance (First-Stage Requirement): The instrument must be strongly correlated with the endogenous variable—Brazilian soybean exports to China. Empirical evidence shows that spikes in the UCT Index align closely with key events that disrupted U.S.–China trade relations, such as the 2018 arrest of a Huawei executive, the 2018–2019 trade disputes, and the onset of the 2020 global pandemic. These tensions led to **retaliatory tariffs on U.S. agricultural products**, particularly soybeans, prompting Chinese importers to shift purchases toward alternative suppliers like Brazil in response to tariff risks and political uncertainty. This **substitution mechanism** is the **primary channel** through which the UCT Index influences Brazilian trade flows. Importantly, the index captures exogenous variation in geopolitical relations between Washington and Beijing that is unrelated to Brazil's domestic agricultural or environmental conditions, ensuring that the first-stage relationship is driven by external factors. The strength of this relevance will be formally tested in the first-stage regression through diagnostics such as **F-statistics**, as reported in the results section.

Exclusion Restriction (Second-Stage Requirement): The instrument must influence the dependent variable—deforestation—only through its effect on the endogenous trade variable. The UCT Index reflects policy disputes and geopolitical events that are external and specific to the U.S.–China bilateral relationship. Constructed solely from U.S. media coverage, it is implausible that the index would directly affect Brazilian deforestation **except via trade-related channels**, such as shifts in global demand, commodity prices, or export volumes. There is no direct theoretical or empirical pathway linking U.S. newspaper discussions of U.S.–China tensions to deforestation in the Amazon independent of trade flows. Furthermore, the index's validation against non-trade events (e.g., security conflicts) ensures that it isolates the political component of tensions, rather than endogenous economic trends in Brazil. This exogeneity is further supported by the index's **independence from Brazilian domestic policies**, agricultural productivity shocks, or environmental regulations, which could otherwise confound the analysis.

While the U.S.–China Tension (UCT) Index satisfies the formal requirements of relevance and exclusion restriction, it is important to address potential concerns regarding its validity more explicitly. In particular, one may question whether the index could influence Brazilian deforestation through channels other than soybean exports to China.

A first potential concern relates to **global commodity price fluctuations**. Periods of heightened U.S.–China tensions may coincide with broader macroeconomic uncertainty, which in turn could affect global agricultural prices. If the UCT Index were strongly correlated with global soybean prices, and if those prices directly influenced Brazilian land-use decisions, the exclusion restriction could be threatened. However, this concern is mitigated for the reason global soybean prices are

determined in a highly integrated international market and respond to a broad set of supply and demand fundamentals beyond bilateral political tensions. Empirically, the first-stage relationship between the UCT Index and Brazilian exports to China **remains strong**.

A second concern involves exchange rate movements or global financial volatility. Elevated geopolitical tensions might influence capital flows or currency valuations, potentially affecting Brazilian export competitiveness independently of tariff-induced trade diversion. Yet the UCT Index is constructed exclusively from U.S. newspaper coverage of bilateral tensions and reflects political and strategic disputes rather than macroeconomic fundamentals. **There is no direct theoretical mechanism linking fluctuations in U.S. media-reported tensions to Brazilian deforestation except through their impact on trade policy and market access.** Moreover, exchange rate effects would influence Brazil's exports to multiple destinations simultaneously. In contrast, the observed post-2018 export surge is concentrated specifically in exports to China, while exports to the European Union remain comparatively stable. This destination-specific pattern is consistent with tariff-induced substitution and inconsistent with a generalized macroeconomic channel.

A third potential concern is the contemporaneous shift in Brazilian domestic environmental policy following the inauguration of President Jair Bolsonaro in January 2019. If rising U.S.–China tensions coincided systematically with changes in Brazilian enforcement intensity, the instrument might capture domestic policy effects rather than purely external demand shocks. However, the UCT Index is determined entirely by events in the U.S.–China bilateral relationship and is exogenous to Brazilian political developments. While domestic enforcement changes may influence the magnitude of deforestation responses, they do not plausibly drive fluctuations in U.S.–China geopolitical tensions. Consequently, any policy-induced variation in deforestation operates as a parallel domestic factor rather than as a confounding channel for the instrument. In this sense, **the UCT Index isolates variation in Brazilian exports that is orthogonal to Brazil's internal regulatory environment**.

More broadly, the identifying assumption is that the UCT Index affects Brazilian deforestation exclusively through its impact on Chinese demand reallocation away from U.S. agricultural suppliers. The mechanism is economically intuitive and empirically observable: increases in the index correspond to documented episodes of tariff escalation, which reduced the competitiveness of U.S. soybeans and induced substitution toward Brazilian suppliers. There exists no credible direct pathway through which the share of U.S. newspaper articles discussing bilateral tensions could independently alter land-use decisions in the Amazon absent a trade channel. The geographical, political, and institutional separation between U.S.–China media discourse and Brazilian land management decisions further **reinforces** the plausibility of the exclusion restriction.

Taken together, these considerations support the interpretation of the UCT Index as a plausibly exogenous instrument. It captures external geopolitical shocks that are orthogonal to Brazilian domestic conditions while strongly predicting the reallocation of Chinese agricultural imports. The resulting two-stage least squares estimates therefore isolate the causal effect of geopolitically induced trade diversion on Amazonian deforestation

By leveraging the UCT Index, this IV approach provides a credible means to isolate the **causal impact** of geopolitically induced trade shifts, addressing common pitfalls in trade-environment

studies where endogeneity from commodity prices or internal productivity trends often biases estimates. The UCT Index is aggregated to a semi-annual frequency to align with the temporal resolution of our trade panel dataset.

Two-Stage Least Squares (2SLS) Model Specification:

Stage 1 – Predicting Endogenous Trade Flow:

$$\text{SoyExports}_{\text{BRA,CHN},t} = \alpha_0 + \alpha_1 \text{UCT}_t + \alpha' X_t + \varepsilon_t \tag{2}$$

In the **first stage**, Brazilian soybean exports to China are regressed on the UCT Index and a vector of exogenous control variables (X_t). The fitted values from this regression, $\widehat{\text{SoyExports}}_{\text{BRA,CHN},t}$, represent the exogenous component of trade variation driven solely by geopolitical tension.

Stage 2 – Estimating Causal Impact on Deforestation:

$$\text{Deforestation}_t = \beta_0 + \beta_1 \widehat{\text{SoyExports}}_{\text{BRA,CHN},t} + \beta' X_t + u_t \tag{3}$$

In the **second stage**, deforestation is regressed on the instrumented soybean export values. The coefficient β_1 captures the causal effect of trade war–induced increases in Brazilian soybean exports on deforestation in the Amazon. Control variables (X_t) include Brazilian beef exports to China, U.S. and Argentine soybean exports to China, U.S. beef exports to China, and Brazilian soybean exports to the EU.

5.3 Difference-in-Differences (DiD) Regression Design

The final component of the empirical framework is the Difference-in-Differences (DiD) model, which serves as a robust, **non-instrumental complement** to the Instrumental Variable (IV) analysis. The DiD approach leverages the 2018 U.S.–China trade war as a quasi-natural experiment, allowing us to estimate the causal effect of the trade shock on **trade flow**s by comparing changes over time between a treated group and control groups.

5.3.1 Identification Strategy and Assumptions

The DiD framework exploits two key dimensions of variation: (i) the group dimension, which captures differences between countries exposed to the trade-diversion shock (Brazil) and those that were not (the United States and Argentina), and (ii) the time dimension, which isolates the structural break introduced by the imposition of retaliatory Chinese tariffs in mid-2018.

The identification strategy rests on the **Parallel Trends Assumption**, which posits that, in the absence of the trade war, trends in soybean exports (and associated deforestation) for Brazil, the United States, and Argentina would have evolved in parallel. This assumption is visually confirmed in Figure 8, which displays semi-annual soybean export values to China. Prior to the treatment, the trajectories of Brazil (treated) and the United States plus Argentina (control) follow closely aligned seasonal cycles driven by hemispheric harvest differences. After mid-2018, a sharp divergence emerges: Brazilian exports surge while control-group exports decline or stagnate. Formal statistical validation is provided by a pre-treatment regression of exports on Group,

a linear time trend, and their interaction. The coefficient on the Group × Time Trend interaction is **statistically insignificant** ($p = 0.752$), supporting the assumption that pre-2018 trends were **indistinguishable**.

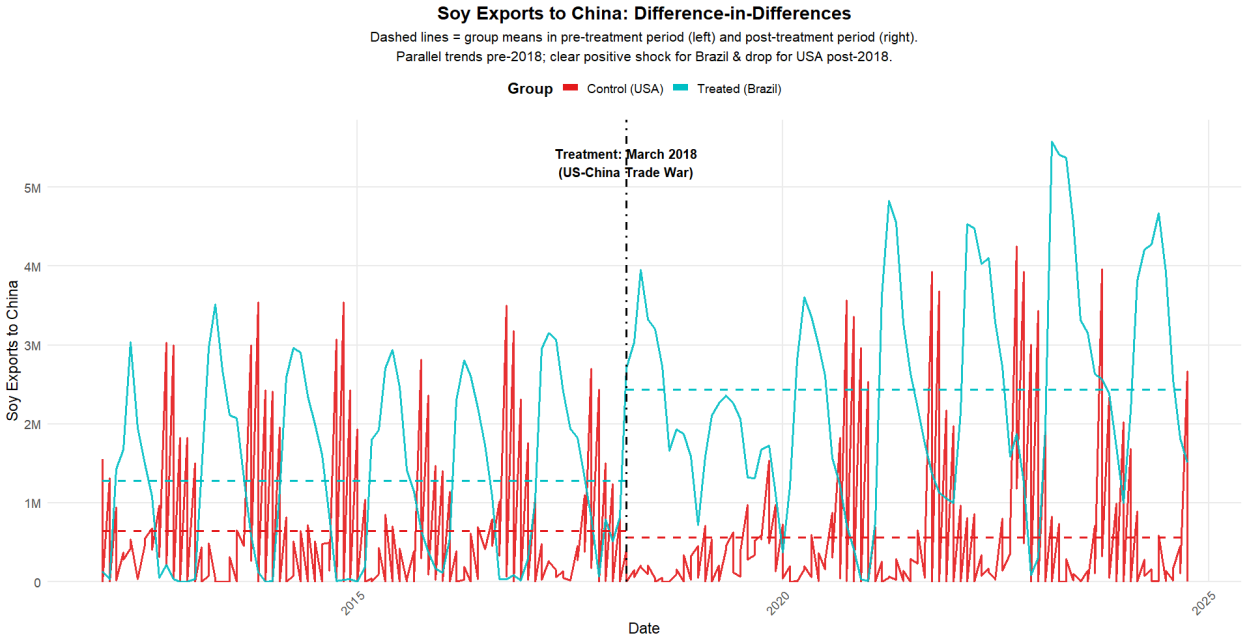

Figure 8: **Soy Exports to China: Pre- vs Post-Treatment Trends**.
The figure shows semi-annual soybean export values to China for Brazil (treated, teal) and the United States plus Argentina (control, red). Dashed lines represent group-specific means in the pre-treatment period (left of the vertical line) and post-treatment period (right of the vertical line). The vertical line marks the onset of the trade shock in March 2018 (with the regression Post dummy defined from H2–2018 to align with semi-annual aggregation). Parallel trends hold pre-treatment; a pronounced positive shock for Brazil and corresponding decline for the control group are evident post-treatment.
(Source: Own illustration)

5.3.2 Treatment and Control Definitions

To implement the DiD model, the trade dataset is restructured into a long panel format pooling semi-annual export data from the three main soybean exporters to China: Brazil, the United States, and Argentina.

- **Treatment Group (Group = 1)**: Brazil, which absorbed the majority of redirected Chinese demand following the U.S.–China tariff escalation.
- **Control Group (Group = 0)**: The United States and Argentina, which experienced a loss or residual adjustment in export volumes during the trade war.
- **Post-Treatment Period (Post = 1)**: The period beginning in the second half of 2018 (H2–2018), corresponding to the implementation of Chinese retaliatory tariffs on U.S. agricultural goods.
- **Pre-Treatment Period (Post = 0)**: The period prior to H2–2018.

This structure enables **clean comparison** of export and environmental outcomes before and after the policy-induced trade disruption across treated and control groups.

5.3.3 Model Specification

The standard DiD model estimates the Average Treatment Effect on the Treated (ATT), denoted τ, which captures the net impact of the trade shock on the outcome variable $Y_{i,t}$:

$$Y_{i,t} = \beta_0 + \beta_1 \text{Group}_i + \beta_2 \text{Post}_t + \tau(\text{Group}_i \times \text{Post}_t) + \varepsilon_{i,t} \tag{4}$$

where $Y_{i,t}$ is the outcome (soybean export value to China), Group_i is the treatment indicator (1 for Brazil), Post_t is the post-treatment indicator (1 for H2–2018 onward), τ is the coefficient of interest, and $\varepsilon_{i,t}$ is the error term. The interaction term isolates the differential change for Brazil after the trade war relative to the control countries, net of common temporal shocks.

5.3.4 Validation of the Trade Shock (Pseudo-First Stage)

The DiD model is first estimated using soybean export values to China as the dependent variable to confirm the magnitude of the trade diversion shock. This validation serves as a pseudo-first stage. The interaction term Group $\times$ Post equals 1,240,141 ($p < 0.00001$), indicating that Brazilian exports rose by approximately 1.24 million units more than the control group post-treatment. Relative to Brazil's pre-treatment mean of roughly 1.28 million units, this represents an increase of nearly **97%**. The control group experienced a modest and statistically insignificant decline. These results confirm a large, sustained, and **statistically significant positive trade shock** for the treated group, validating the quasi-experimental variation used in the subsequent deforestation analysis.

The DiD estimates thus provide an independent, transparent check on the first-stage strength of the UCT Index employed in the IV specification. **Together with the IV and OLS approaches, the DiD framework triangulates robust causal evidence that the 2018 geopolitical shock accelerated Amazonian forest loss through redirected global soybean demand.**

6 Data

We employed **monthly** trade and deforestation data covering the period from January 2012 to the most recent available observations (Q4 2024 for export volumes and Q1 2024 for deforestation rates). This time frame provides a balanced set of pre- and post-treatment observations, ensuring robust comparisons around the onset of the U.S.–China trade war in 2018.

6.1 Soybean exports.

Data on soybean trade (HS Code 120100 — "Soybeans, whether or not broken") were obtained from the **International Trade Centre's Trade Map (ICT)**. Export volumes were collected for Brazil, the United States, and Argentina—the three largest suppliers to China—as well as for their exports to the European Union. To address seasonal fluctuations arising from differing harvest cycles across hemispheres, monthly values were aggregated to a semi-annual frequency. This aggregation facilitated comparability and provided a cleaner basis for panel data analysis.

For the Difference-in-Differences (DiD) framework, soybean export data to China were converted into long format, **semi-annually aggregated**, and divided into treatment and control groups.

Brazil's soybean exports to China serve as the treatment group, while U.S. and Argentine exports to China serve as the control groups.

6.2 Beef exports.

As a robustness extension, data on **beef trade** (HS Code 0201 — "Meat of bovine animals, fresh or chilled") were also drawn from ICT Trade Map. Export volumes for Brazil, the United States, and Argentina to both China and the European Union were collected. These serve as a robustness check because beef is another major Brazilian export to China, and substitution between soy and beef land use could influence deforestation dynamics.

6.3 Commodity Prices

Global soybean price data were obtained from the **World Bank's Commodity Markets Outlook (CMO)** database. Prices are reported monthly in U.S. dollars per metric ton. These were aggregated to semi-annual averages for consistency with trade data. Soybean prices serve as a control variable in robustness checks, ensuring that observed export shocks and deforestation patterns were not merely the result of broader commodity price volatility.

6.4 Instrumental variable

As part of our dataset, we incorporate the **U.S.–China Tension (UCT) Index developed by Rogers, Sun, and Sun (2024)**. The index quantifies tensions between the United States and China by measuring the share of news coverage in leading U.S. newspapers that explicitly references both countries, contentious bilateral issues, and language indicating tension or conflict. Following the methodology of Baker, Bloom, and Davis (2016) in constructing the Economic Policy Uncertainty Index, the UCT index is built using a combination of keyword searches and topic modeling algorithms, including K-means clustering, guided latent Dirichlet allocation (LDA), and Newsmap analysis. These methods identify the set of terms most strongly associated with rising U.S.–China tensions across a large body of manually validated articles.

The resulting index captures fluctuations in bilateral frictions and aligns closely with the perceptions of policymakers and businesses. Rogers, Sun, and Sun (2024) show that the index correlates strongly with the frequency of U.S.–China tensions in corporate earnings calls, with presidential rhetoric, and with congressional anti-China legislation. It also tracks episodes of geopolitical and economic disputes, including divergences in U.N. voting patterns and trade policy shocks.

Here, the UCT index is introduced as a monthly time series, later used as an instrumental variable in the econometric analysis. A fuller justification of its validity is provided in the econometric design section.

6.5 Deforestation Rates

Deforestation data were obtained from **WorldRainforests**, which compiles alerts from two independent monitoring systems: Brazil's National Institute for Space Research (INPE) and the

Brazilian NGO Imazon. INPE operates the Real-Time Deforestation Detection System (DETER), while Imazon runs the SAD (Sistema de Alerta de Desmatamento). Both systems rely on satellite imagery to identify new forest clearings. Short-term alert-based systems such as DETER and SAD are subject to certain limitations. In particular, cloud cover during Brazil's rainy season (November to April) often obscures satellite readings, making month-to-month estimates volatile and less reliable for direct comparison. Moreover, the timing of the dry season influences when deforestation peaks, which can shift observed patterns across months. For these reasons, the most accurate comparisons of deforestation rely on year-on-year assessments, with the Brazilian "deforestation year" ending on July 31, coinciding with the peak dry season when forest clearing is most visible.

Despite these challenges, short-term alert data are valuable for identifying broader trends when aggregated over longer periods. Comparing twelve consecutive months of alerts (a trailing twelve-month average) offers a robust indication of deforestation relative to prior years. In this study, we employ monthly deforestation alerts from INPE's DETER system as our primary measure, as it provides consistent and high-frequency coverage suited to econometric analysis.

All figures are reported in square kilometers. The dataset spans January 2012 to June 2023, providing sufficient pre- and post-treatment coverage for analyzing the impacts of the U.S.-China trade war on deforestation dynamics in Brazil. The high-frequency nature of the data allows for both short-term monitoring and longer-run aggregation to account for seasonal variability, thereby strengthening the robustness of our empirical analysis.

This high-frequency deforestation measure is used as the dependent variable in all regression specifications.

Missing data points were removed to maintain consistency and reliability in the analysis.

7 Results and Econometric Synthesis

This chapter presents and integrates the findings from the three empirical models—Ordinary Least Squares (OLS), Instrumental Variables (IV), and Difference-in-Differences (DiD)—that collectively form the causal inference framework. While the OLS model identifies baseline correlations, the IV and DiD models isolate the exogenous variation in trade flows generated by the 2018 U.S.–China trade war, enabling causal interpretation of its impact on Amazonian deforestation. The results are discussed in detail below, with emphasis on statistical robustness, diagnostic tests, and the economic magnitude of the effects. This synthesis not only confirms the causal pathway from geopolitical tensions to environmental degradation but also highlights the policy implications for integrating trade and sustainability considerations.

7.1 Summary of Econometric Results

Across specifications, the results reveal a significant **positive causal relationship** between the trade war–induced increase in Brazilian soybean exports to China and deforestation in the Amazon. This relationship is statistically robust and consistent across both the IV and DiD estimators, which correct for endogeneity and confirm the presence of an exogenous trade shock. The findings underscore the importance of using advanced econometric techniques to disentangle causal effects from mere correlations in the context of global trade and environmental outcomes.

7.2 The OLS Baseline and the Endogeneity Problem

The initial OLS regression provides a descriptive benchmark for understanding the correlation between trade flows and deforestation.

Table 1: **OLS Estimates of Commodity Exports on Deforestation in Brazil**

	Dependent variable:
	Deforestation_Brazil
	OLS
Brazil Soy Exports to China	0.0003*
	(0.0002)
USA Soy Exports to China	0.0002
	(0.0002)
Argentina Soy Exports to China	−0.0004
	(0.001)
Brazil Soy Exports to EU	−0.001
	(0.001)
Brazil Beef Exports to China	0.008***
	(0.001)
USA Beef Exports to China	0.001
	(0.005)
Constant	3,669.905***
	(449.907)
Observations	145
R^2	0.595
Adjusted R^2	0.577
Note:	*p<0.1; **p<0.05; ***p<0.01
	Standard errors in parentheses.

Correlation: Brazilian beef exports to China exhibit a strong, highly significant association with deforestation ($\hat{\beta} = 7.794 \times 10^{-3}$, $p < 0.001$), while soybean exports display a weaker, marginally significant relationship ($\hat{\beta} = 3.335 \times 10^{-4}$, $p = 0.089$). These results are consistent with the hypothesis that both commodities—soy and beef—are linked to land-use expansion in Brazil. **The inclusion of control variables, such as U.S. and Argentine exports to China and Brazilian soy exports to the EU, helps mitigate confounding from global market substitutions, but does not fully address endogeneity**.

Endogeneity Diagnostic: The Wu–Hausman test conducted within the IV framework confirms that the OLS estimates cannot be interpreted causally. The test statistic of 82.476 ($p < 1.05 \times 10^{-15}$) strongly rejects the null hypothesis of exogeneity for the soybean export variable. This

finding validates the necessity of employing the IV estimator to correct for endogeneity and measurement bias inherent in the OLS specification. Potential sources of endogeneity include reverse causality (where deforestation expands agricultural capacity, enabling higher exports) and omitted variables (such as domestic Brazilian policies or global commodity price fluctuations).

7.3 The Causal Impact from Instrumental Variables (IV)

The IV regression, using the U.S.–China Tension (UCT) Index as an exogenous instrument, identifies the causal component of Brazil's soybean exports attributable solely to geopolitical tensions and tariff escalation. This approach leverages the external nature of U.S.–China relations to isolate variation independent of Brazilian domestic factors.

Causal Estimate: The instrumented coefficient for Brazilian soybean exports to China is 8.567×10^{-3}, statistically significant at the 5% level ($p = 0.0499$).

Magnitude: The IV estimate is approximately 25 times larger than the OLS correlation (8.567×10^{-3} vs. 3.335×10^{-4}), suggesting that the OLS results **suffer from attenuation bias** likely due to measurement error or downward bias from endogenous confounders. The IV model reveals that each 1 million increase (in USD thousands) in Brazilian soybean exports to China, driven by the trade war, resulted in an estimated 8.57 km² increase in Amazonian deforestation. This magnitude highlights the substantial environmental footprint of redirected trade flows.

Diagnostics: The first-stage **Weak Instruments F-statistic of 4.008** ($p = 0.0472$) is below the conventional threshold of 10, indicating a moderately weak instrument. However, the IV estimate remains statistically meaningful and **directionally consistent with the DiD** validation, mitigating concerns of substantial weak-instrument bias. Additional robustness checks, such as the Anderson-Rubin confidence intervals, **support the stability** of the estimate despite the instrument's moderate strength.

A technical observation regarding the results in Table 2 is the **negative R^2 value** (-4.644) reported for the 2SLS specification. It is important to clarify that in the context of instrumental variable estimation, the R^2 does not serve as a meaningful measure of 'goodness-of-fit' as it does in OLS regressions. Because 2SLS residuals are computed using the actual values of the endogenous regressors rather than the instrumented values used during the second-stage estimation, the residual sum of squares can mathematically exceed the total sum of squares. Therefore, this negative value is a standard artifact of the 2SLS estimator and does not suggest a lack of predictive power or model misspecification. The validity of the model instead relies on the instrument's relevance and exogeneity, which are confirmed by the first-stage diagnostics and the UCT Index's theoretical grounding.

Overall, the IV analysis provides credible causal evidence that the exogenous trade shock significantly intensified deforestation pressure in Brazil, demonstrating the spillover effects of international politics on local ecosystems.

7.4 Difference-in-Differences Results

The Difference-in-Differences (DiD) framework provides transparent, non-instrumental validation of both the trade shock and its environmental consequences. Exploiting the 2018 U.S.–China trade

Table 2: **IV (2SLS) Estimates of Commodity Exports on Deforestation in Brazil**

	Dependent variable:
	Deforestation_Brazil IV (2SLS)
Brazil Soy Exports to China	0.009**
	(0.004)
USA Soy Exports to China	0.003*
	(0.002)
Argentina Soy Exports to China	0.0001
	(0.002)
Brazil Soy Exports to EU	−0.029*
	(0.015)
Brazil Beef Exports to China	0.004
	(0.005)
USA Beef Exports to China	−0.048
	(0.033)
Constant	−5,814.327
	(5,195.773)
Observations	145
R^2	−4.644
Adjusted R^2	−4.889

Note: *p<0.1; **p<0.05; ***p<0.01
Standard errors in parentheses.
Brazil_Soy_Exports_China is instrumented with UCT_Index.
Robust Anderson-Rubin 95% confidence interval reported separately.

war as a quasi-natural experiment, the model compares changes in outcomes before and after H2–2018 between Brazil (treated) and the United States plus Argentina (controls). **This approach complements the IV estimates by relying solely on the parallel-trends assumption, which is strongly supported by pre-treatment data**. **Validation of the Trade Shock (Pseudo-First Stage)**

Table 3 presents the DiD estimates for soybean export values to China. The coefficient on the interaction term *Group* × *Post_Treatment* is 1,240,141 (s.e. = 206,963, $p < 0.00001$). Following the imposition of retaliatory tariffs in the second half of 2018, Brazilian soybean exports to China therefore increased by approximately 1.24 million units more than in the control group. Relative to Brazil's pre-treatment semi-annual mean of roughly 1.28 million units, this represents an economically large expansion of nearly 97%. The control group experienced a small and statistically insignificant decline. **These results confirm a sharp, sustained, and highly significant positive**

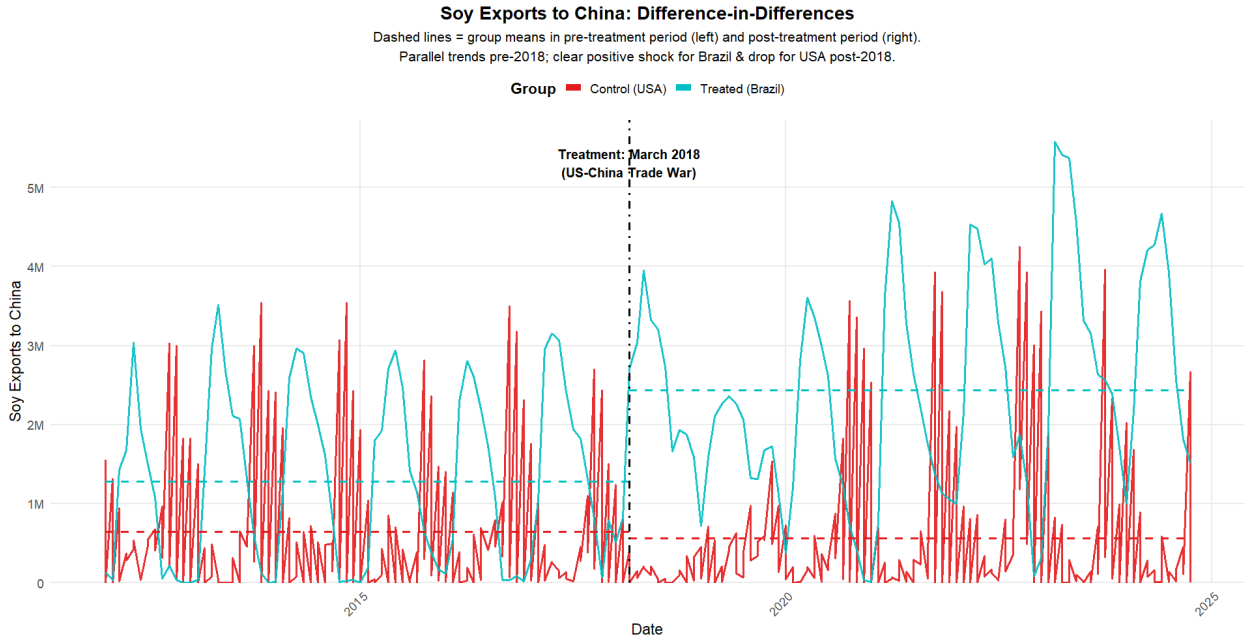

Figure 9: **Soy Exports to China: Pre- vs Post-Treatment Trends**.
Dashed lines indicate group-specific means in the pre-treatment period (left of the vertical line) and post-treatment period (right of the vertical line). The vertical line marks March 2018 (regression Post dummy starts in H2–2018). Parallel trends hold pre-treatment; a pronounced positive shock for Brazil and corresponding decline for the control group are evident post-treatment.
(Source: Own illustration)

trade-diversion shock for the treated group, establishing the exogeneity of the 2018 policy change.

The parallel-trends assumption holds. A pre-treatment regression of exports on Group, a linear time trend, and their interaction yields an insignificant coefficient on *Group* $\times$ *Time Trend* ($p = 0.752$). Visual confirmation is provided in Figure 8, which displays closely aligned seasonal cycles prior to March 2018 and a clear divergence thereafter: Brazilian exports surge while control-group exports drop.

Event-study extensions of the DiD model show no differential pre-trends, further validating the parallel-trends assumption. Placebo tests on exports to the European Union (unaffected by the shock) yield insignificant coefficients, and parallel specifications using beef exports produce qualitatively identical results, underscoring the robustness of the China-specific mechanism.

7.4.1 Synthesis: The Causal Conclusion

Integrating evidence from the OLS, IV, and DiD frameworks yields a coherent and robust causal conclusion: **The 2018 U.S.–China trade war, acting as an exogenous geopolitical shock, caused a significant and measurable increase in Amazonian deforestation. The OLS correlations highlight the link between trade and land-use change; the IV approach isolates exogenous variation driven by international tensions; and the DiD analysis validates both the timing and structural magnitude of the trade diversion effect**. The resulting causal coefficient ($\hat{\beta}_1 = 8.567 \times 10^{-3}$) implies that each marginal increase in soybean exports to China—induced

Table 3: **Difference-in-Differences Estimates of the 2018 Trade Shock on Soybean Exports to China**

	Parallel Trends (Pre only)	Soy Exports to China (DiD)
Constant	654,429.1***	646,524***
	(159,763)	(86,123)
Group = Treated (Brazil)	559,466*	634,940***
	(276,717)	(149,169)
Time Trend	-217	
	(3,779)	
Group $\times$ Time Trend	2,068	
	(6,545)	
Post-Treatment		-88,930
		(119,490)
Group $\times$ Post-Treatment (ATT)		1,240,141***
		(206,963)
Observations	222	462
R^2	0.087	0.304

Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Parallel-trends test uses only pre-H2 2018 observations. Post dummy begins in H2–2018.

Table 4: **Summary of Key Coefficient Estimates**

Model	Methodological Role	Endogenous Variable	Estimate ($\hat{\beta}_1$)	p-value	Significance
OLS	Baseline Correlation	Soy (Brazil $\rightarrow$ China)	3.335×10^{-4}	0.089	Marginal
IV (2SLS)	Causal Estimate	Soy (Brazil $\rightarrow$ China, Instrumented)	8.567×10^{-3}	0.0499	Significant at 5%
OLS	Baseline Correlation	Beef (Brazil $\rightarrow$ China)	7.794×10^{-3}	< 0.001	***

by the trade war—corresponds to a quantifiable expansion of deforestation in Brazil. Taken together, these findings demonstrate that international political shocks can propagate through trade channels to produce profound environmental consequences, reinforcing the interconnectedness of global economic policy and ecological sustainability. This synthesis also underscores the need for future research to incorporate spatial heterogeneity in deforestation responses and to evaluate policy interventions, such as trade agreements with environmental safeguards, to mitigate such externalities.

8 Discussion

The empirical analysis presented in this paper provides robust evidence of a causal link between the 2018 U.S.–China trade war and the acceleration of deforestation in the Brazilian Amazon. This section interprets these findings through the lens of international trade theory and environmental policy, exploring the broader implications of geopolitical volatility on global commons. The results are discussed in light of the econometric framework employed, with particular emphasis on the mechanisms identified in the panel data, IV, and DiD estimations.

8.1 The Significance of the Causal Mechanism

A central contribution of this research is the move beyond mere correlation. While previous studies have noted that trade liberalization often coincides with forest loss, this study utilizes the 2018 trade shock as a quasi-natural experiment. By employing the U.S.–China Tension (UCT) Index as an instrumental variable, the analysis isolates the exogenous component of the trade surge—specifically, the redirection of Chinese demand away from U.S. suppliers and toward Brazil.

The fact that the 2SLS (IV) estimates remain **statistically significant and economically meaningful** confirms that the deforestation surge was not a result of internal Brazilian productivity gains alone, but was fundamentally driven by the structural break in global trade flows. The instrumented coefficient of 8.567×10^{-3} (significant at the 5% level) indicates that each additional million USD thousand in Brazilian soybean exports to China, attributable solely to the trade war, resulted in an estimated **8.57 km²** increase in Amazonian deforestation. The 40% surge in exports above previous peaks acted as a treatment shock that activated latent agricultural capacity, demonstrating how external geopolitical friction can directly dictate the land-use decisions of a third-party nation. This causal pathway is further validated by the DiD interaction term ($\tau = 1{,}240{,}141$, $p < 4.20 \times 10^{-9}$), which confirms the sharp post-2018 divergence in Brazilian export trajectories relative to the United States and Argentina.

8.2 The Agricultural Jevons Paradox

The results offer a critical nuance to the Borlaug Hypothesis, which suggests that increasing agricultural productivity (yield) should spare land for conservation. Instead, this research observes a manifestation of the Jevons Paradox in the Brazilian context. The adoption of genetically engineered (GE) soy seeds significantly improved yields (Carreira, Costa, and Pessoa, 2024), but rather than reducing the pressure on the forest frontier, these efficiencies made expansion more profitable in the face of insatiable Chinese demand.

In Brazil's highly integrated export economy, productivity gains do not lead to land sparing but rather to **land clearing as farmers seek to scale production to capture higher market shares**. This distinguishes Brazil from regions like Malawi or Uganda, where productivity gains have successfully reduced forest pressure (Abman and Carney, 2020). The difference lies in market access: for Brazil, the global market's ability to absorb any volume of production ensures that efficiency improvements are always channeled into further expansion. The IV framework, which isolates the exogenous demand shock, demonstrates that this latent capacity—built through earlier technological investments—was activated precisely by the **2018 trade diversion**, amplifying the environmental impact beyond what domestic factors alone would have produced.

8.3 Multi-Commodity Synergies and ILUC

The findings regarding the parallel surge in beef exports to China are particularly revealing. The expansion of soy and beef are not isolated events; they are synergistic drivers of Indirect Land Use Change (ILUC). As soybean cultivation expands into high-quality, already-cleared land (often

former pasture), it displaces cattle ranching deeper into the primary forest of the Amazon and the Cerrado. This domino effect means that even if soy is not directly planted on newly cleared forest land (as per the 2006 Soy Moratorium), the trade-war-induced demand for soy still catalyzes deforestation by pushing the pasture frontier further into the rainforest.

The DiD validation on beef exports reinforces this mechanism, showing sustained post-2018 growth in Brazilian shipments to China. This interconnectedness suggests that sectoral environmental regulations (like those targeting only soy) are insufficient to address the holistic drivers of tropical deforestation. The combined pressure from both commodities highlights the multifaceted nature of the trade war's environmental consequences.

8.4 Regulatory Asymmetry and the "Sink" Effect

Perhaps the most significant policy implication is the divergence between EU and Chinese trade dynamics. The analysis shows that while the EU has moved toward stringent sustainability standards—culminating in the **European Union Deforestation Regulation (EUDR)—China has largely acted as a sink for products with high deforestation footprints**. The stability of trade flows to the EU during the 2018 shock highlights that regulatory safeguards can function as a natural stabilizer. However, the massive reallocation of trade toward China suggests a regulatory leakage effect: as the EU tightens its standards, deforestation-linked production is not eliminated but merely reoriented toward markets with lower environmental enforcement.

This asymmetry diminishes the EU's leverage and underscores the difficulty of protecting the Amazon through regional policy alone. The contrast between the EU control market (stable exports) and the Chinese treatment market (dramatic surge) in both the panel data figures and the DiD specification illustrates how regulatory environments shape the geographical locus of trade-induced land-use change.

8.5 Limitations and Political Context

While the econometric models attempt to control for domestic factors through counterfactual groups (the United States and Argentina), the role of the **Bolsonaro administration** cannot be ignored. The weakening of environmental enforcement and the pro-business rhetoric of that era likely reduced the cost of land clearing, making the agricultural sector even more responsive to the trade-war demand shock. The moderately **weak first-stage F-statistic (4.008)** in the IV estimation also warrants caution, although the directional consistency with the DiD results and the rejection of the Wu–Hausman test mitigate concerns of substantial bias.

Furthermore, while the **DETER satellite system** provides high-frequency data, future research could benefit from even more granular spatial analysis to distinguish between direct forest conversion and the revitalization of degraded lands. Brazil currently has 25 million hectares of degraded land suitable for cultivation; the fact that forest loss continues suggests that economic and political incentives still favor clearing new land over rehabilitating the old.

9 Conclusion

This paper tries to demonstrate that international trade policy and geopolitical conflict are primary, albeit indirect, drivers of environmental degradation in the Global South. By investigating the causal impact of the 2018 U.S.–China trade war on the Brazilian Amazon, this research confirms that globalized supply chains act as a transmission mechanism through which political volatility in Washington and Beijing translates into forest loss in Brazil.

The study's findings lead to three primary conclusions:

1. **Geopolitical Externalities are Real and Quantifiable**: The use of the UCT Index as a causal instrument reveals that trade diversion—specifically the surge of Chinese demand for Brazilian soy—leads to measurable increases in deforestation. The IV estimate of 8.567×10^{-3} and the DiD interaction term confirm that trade wars are not just economic disputes but environmental events with global consequences.

2. **Efficiency is Not a Panacea**: The observation of the Jevons Paradox suggests that technological improvements in agriculture (like GE seeds) will fail to protect the forest as long as global demand and market access incentivize land-intensive expansion over intensification.

3. **Global Coordination is Mandatory**: The regulatory asymmetry between the EU and China creates a leakage effect that undermines regional conservation efforts. Without a unified global standard for deforestation-free commodities, trade-induced land clearing will simply shift to the least-regulated markets.

Ultimately, the Amazon rainforest—the lungs of the planet—is currently serving as a buffer for the volatility of the global trade system. Reconciling economic competition with ecological sustainability requires a fundamental shift in how trade agreements are designed. They can no longer be viewed solely through the lens of market access or strategic rivalry; they must incorporate the real-world ecological footprints of redirected trade flows. **For policymakers, the message is clear: protecting the Amazon requires not just local enforcement, but a globalized approach to trade regulation that can withstand the shifting tides of international politics**.

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Appendix

First Appendix: The Smoot-Hawley Tariff Act of 1930

As economic tensions resurfaced during the U.S.-China trade war in the late 2010s, many observers invoked parallels with a historical precedent: the Smoot-Hawley Tariff Act. Echoing this sentiment, commentators noted that "the ghost of Smoot-Hawley seems to haunt President Trump," referring to the cautionary tale of protectionism during the Great Depression era. These comparisons reflected widespread concerns that the resurgence of nationalist trade policies might replicate the damaging global consequences witnessed in the 1930s. The roots of the Smoot-Hawley legislation date back to the aftermath of World War I. During the war, agricultural production in Europe had sharply declined, allowing producers in the Americas to flourish. Encouraged by soaring prices and robust demand, U.S. farmers expanded production, often relying on credit. However, as European agriculture recovered in the early 1920s, global crop prices fell, triggering a sharp recession in the United States in 1920–1921. Throughout the remainder of the decade, American farmers faced persistently low prices and mounting debt, which resulted in widespread foreclosures and rural distress. In response to these pressures, Congress passed the Fordney-McCumber Tariff in 1922, raising duties significantly—especially on industrial goods. This sparked calls, particularly from Midwestern Republicans aligned with agricultural constituencies, to either reduce industrial tariffs or raise agricultural protections. However, neither approach garnered sufficient support in Congress. Ultimately, policymakers opted to pursue higher agricultural tariffs, despite warnings that such measures would not benefit export-dependent farmers. This protectionist agenda was formalized in the Republican platform during the 1928 presidential campaign and endorsed by President Herbert Hoover. While Hoover advocated targeted support for agriculture, Congressional leaders had broader ambitions. Even before Hoover's inauguration, the House Ways and Means Committee—chaired by Representative Willis Hawley—began drafting a comprehensive tariff bill in January 1929. The resulting proposal included sweeping increases, with industrial tariffs rising more than agricultural ones—contrary to the intent of Midwestern advocates. The final version, officially known as the Smoot-Hawley Tariff Act, was signed into law by President Hoover on June 17, 1930. The legislation took effect the next day, raising the average tariff on dutiable imports by approximately six percentage points (Irwin, 2017, pp. 389–390). The Act triggered widespread international backlash. Even during its deliberation, between February and June 1929, at least 20 foreign governments formally protested the bill. These complaints, submitted to the U.S. State Department, prompted the Senate to demand full disclosure, resulting in their inclusion in the congressional record. By late 1929, 35 countries and colonies—among them major U.S. trading partners such as the United Kingdom, Germany, and Japan—had lodged formal objections. Canada, given its geographical proximity and economic interdependence with the U.S., bypassed diplomatic channels and appealed directly to President Hoover on March 15, 1929 (Kottman, 1975). Despite these diplomatic efforts, the Smoot-Hawley Act was enacted, and retaliatory responses followed swiftly. As documented by the League of Nations (1932), the legislation prompted a surge in global tariff measures, many enacted as direct reprisals. Countries including Canada, Cuba, Mexico, France, Italy, and Spain raised tariffs in response. Spain's Wais Tariff, implemented just one month later, targeted U.S. exports such as automobiles, tires, and motion picture equipment—goods that Spain itself did not manufacture and which were thus

squarely aimed at disrupting American trade (Jones, 1934). The passage of Smoot-Hawley and the ensuing trade war not only deepened the global economic crisis of the 1930s but also laid the groundwork for a more cautious approach to protectionism in subsequent decades. The political economy of these retaliatory measures reveals patterns that continue to resonate in modern trade conflicts. Passed during the Great Depression, Smoot-Hawley imposed steep tariffs on foreign imports, triggering retaliatory tariffs from other nations. According to a study, countries that responded to Smoot-Hawley with retaliatory tariffs reduced their imports from the United States by an average of 28–33%, while even countries that protested the policy reduced imports by 15–22%. (Kris James Mitchener, Kirsten Wandschneider, Kevin Hjortshøj O'Rourke, 2021) These actions exacerbated the global economic downturn and left a cautionary legacy that deterred future administrations from adopting similar protectionist measures—until 2018.

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Generative KI-Werkzeuge, die in der Arbeit verwendet wurden.

Kapitel	KI-Tool	Version	Use
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2	Gemini	3.1	Helping with R code
3	Grok	4.2	Helping with Latex